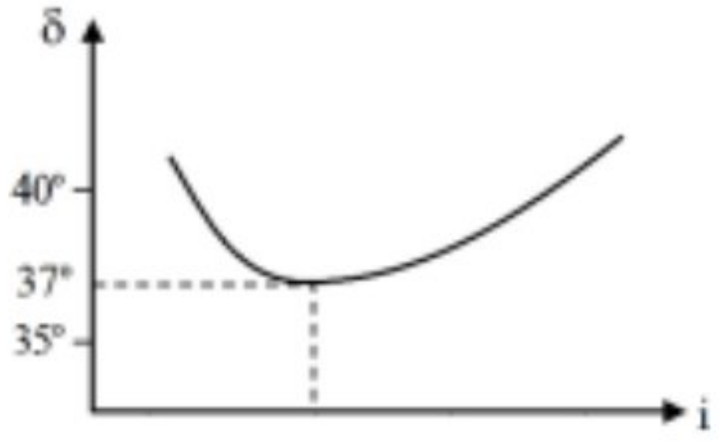
PHYSICS

1) To determine the refractive index of material of glass prism, graph between angle of deviation versus angle of incidence observed in lab is as shown in figure. If angle of prism is 60°, then refractive index of prism will be :



[given, sin(48.5°) = 3/4]

(1) 1.6

(2) 1.5

(3) 1.55

(4) 1.7

2) A wall has two layers A and B each made of different materials. The thickness of both the layers is same. The thermal conductivity of A, $K_A = 3K_B$. The temperature difference across the wall is 20°C in steady state, then :-

(1) The temperature difference across A is 15°C

(2) Rate of heat transfer across A is more than across B

(3) Temperature difference across B is 10°C

(4) Temperature difference across A is 5°C

3) Molar specific heat capacity C :-
(Symbols have their usual meaning)

(1) Can be represented by $\frac{Q}{\mu \Delta T}$

(2) Depends on nature of substance

(3) Depends on temperature of substance

(4) All of the above

4) **Assertion (A) :-** Blue star is at high temperature than red star.

Reason (R) :- Wien's displacement law states that $T \propto \frac{1}{\lambda_m}$.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

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(4) Both **(A)** and **(R)** are correct and **(R)** is correct explanation of **(A)**

5) The temperature at which the average translational kinetic energy of a molecule is equal to the energy gained by an electron in accelerating from rest, through a potential difference of 1 volt is :-

(1) 4.6×10^3 K

(2) 11.6×10^3 K

(3) 23.2×10^3 K

(4) 7.7×10^3 K

6) The coefficients of apparent volumetric expansion of a liquid, when determined using two different vessels A and B are y_1 and y_2 respectively. If the coefficient of linear expansion of the vessel A is α, the coefficient of linear expansion of the vessel B will be :-

(1) $\frac{\alpha y_1 y_2}{y_1 + y_2}$

(2) $\frac{y_1 - y_2 + \alpha}{3}$

(3) $\frac{y_1 - y_2}{2\alpha}$

(4) $\frac{y_1 - y_2}{3} + \alpha$

7) An ideal gas follows VT^2 = constant. Its volume expansion coefficient will be :-

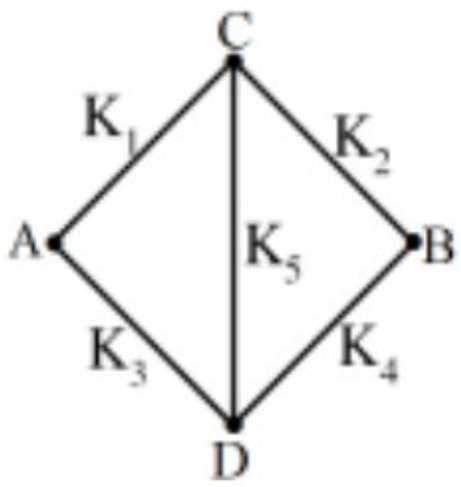
(1) $\frac{2}{T}$

(2) $-\frac{2}{T}$

(3) $\frac{3}{T}$

(4) $\frac{-3}{T}$

8) Five rods of same dimensions are arranged as shown in figure. They have thermal conductivities K_1, K_2, K_3, K_4 and K_5. When points A and B are maintained at different temperatures, no heat flows through the central rod CD if :-



(1) $K_1K_4 = K_2K_3$

(2) $K_1K_2^2 = K_3K_4^2$

(3) $\frac{K_1}{K_4} = \frac{K_2}{K_3}$

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(4) $K_1K_2 = K_3K_4$

9) A gas is heated at a constant pressure. The fraction of heat supplied used for external work is :-

(1) $\frac{1}{\gamma}$

(2) $1-\frac{1}{\gamma}$

(3) $\gamma - 1$

(4) $1-\frac{1}{\gamma^2}$

10) The water equivalent of 120 gm of aluminium (specific heat 0.2 Cal/gm-°C) is :-

(1) 40 gm

(2) 4 gm

(3) 24 gm

(4) None of these

11) A body x at an initial temperature 100°C and another body y at an initial temp. 0°C are placed in an evacuated enclosure, the walls of which are maintained at 10°C. Which one of the following statements is related to prevost theory.

(1) x emits but does not absorb heat

(2) y absorbs but does not emit heat

(3) The final temp. of the bodies will be the mean of their initial temperatures

(4) The walls of the enclosure radiate heat to both x & y

12) The ratio of the densities of the two bodies is 3 : 4 and the ratio of specific heats is 4 : 3. The ratio of their thermal capacities for unit volume is :-

(1) 1 : 1

(2) 4 : 3

(3) 3 : 4

(4) 16 : 9

13) Velocity of four different particles are given as 2V, 10V, 11V and $5\sqrt{3}V$, then the rms velocity of the sample is :-

(1) $5\sqrt{3}V$

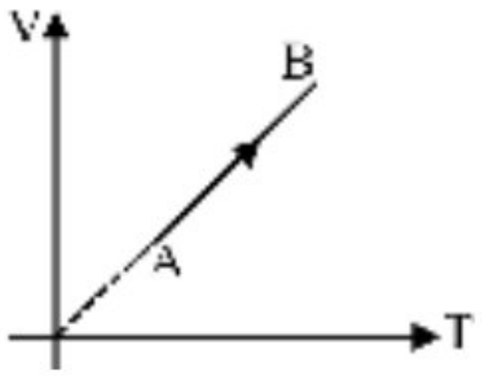
(2) 3V

(3) $3\sqrt{5}V$

(4) 5V

14) An ideal monoatomic gas undergoes the process AB as shown in figure. If the heat supplied and the work done in the process are ΔQ and ΔW respectively. The ratio ΔQ : ΔW is :-

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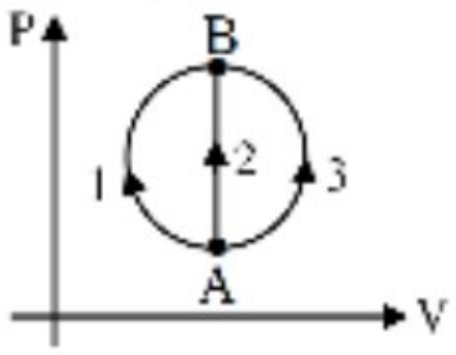
(1) 2.5

(2) 1.67

(3) 1.25

(4) 0.40

15) A gas undergoes A to B through three different processes 1, 2 and 3 as shown in figure. The heat supplied to the gas is Q_1, Q_2 and Q_3 respectively then :-



(1) $Q_1 = Q_2 = Q_3$

(2) $Q_1 < Q_2 < Q_3$

(3) $Q_1 > Q_2 > Q_3$

(4) $Q_1 = Q_3 > Q_2$

16) A solid object of mass 'm' is made from a material of specific heat capacity 'C' and latent heat of fusion 'L'. It enters earth's atmosphere from outer space, its temperature being below its melting point by ΔT. Because of atmospheric friction it absorbs energy at constant rate 'R' the time in which solid becomes completely molten is :

(1) $\frac{m(C+L)\Delta T}{R}$

(2) $\frac{m(C\Delta T+L)}{R}$

(3) $m\,(C+L)\,\Delta TR$

(4) $\frac{R}{M\Delta T(C+L)}$

17) Power radiated by black body is P_0 and the wavelength corresponding to maximum energy is λ_0. On changing the temperature of the black body, it was observed that the power radiated becomes $\frac{256}{81}P_0$. The change in wavelength corresponding to maximum energy will be.

(1) $\frac{+\lambda_0}{4}$

(2) $\frac{+\lambda_0}{2}$

(3) $\frac{-\lambda_0}{4}$

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(4) $\frac{-\lambda_0}{2}$

18) The earth is getting energy from the sun whose surface temperature is T_s and radius is R. Let the radius of the earth is r and the distance from the sun be d. Assume the earth and the sun both behave as perfect black bodies and the earth is in thermal equilibrium at a constant temperature T_e. Therefore, the temperature T_s of the sun is xT_e where x is

(1) $\sqrt{\frac{2d}{R}}$

(2) $\sqrt{\frac{2R}{r}}$

(3) $\sqrt{\frac{4d}{r}}$

(4) $\frac{d}{r}$

19) The molar heat capacity of a certain substance varies with temperature according to given equation

$C = 4T \times 10^{-3} JK^{-1}mol^{-1}$

(T is absolute temperature)

The heat necessary to change the temperature of 2 mol of substance from 100 K to 200 K :-

(1) 100 J

(2) 120 J

(3) 140 J

(4) 80 J

20) A liquid of refractive index 1.5 is poured into a cylindrical jar of radius 20cm upto a height of 20cm. A small bulb is lighted at the centre of the bottom of the jar. Find the area of the liquid surface through which the light of the bulb passes into air:-

(1) 2000 cm^2

(2) 500 cm^2

(3) 1500 cm^2

(4) 1004.6 cm^2

21) Absolute refractive indices of glass and water are $\frac{3}{2}$ and $\frac{4}{3}$. The ratio of velocity of light in glass and water will be :-

(1) 4 : 3

(2) 8 : 7

(3) 8 : 9

(4) 3 : 4

22) A boy stands straight infront of a mirror at a distance of 30 cm away from it. He sees his erect image whose height is $1/5^{th}$ of his real height. The mirror he is using is :-

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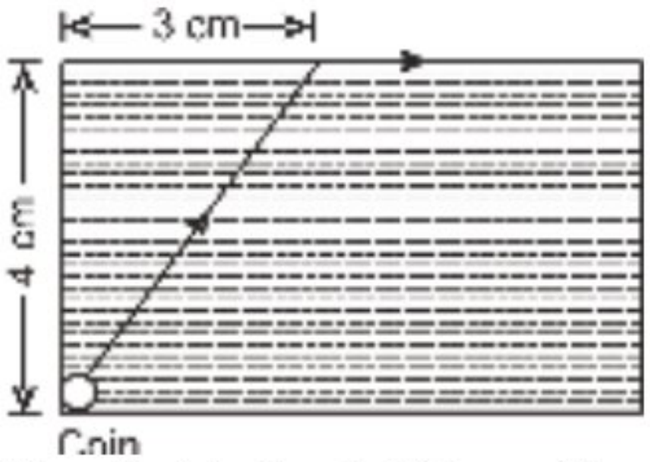
(1) Plane mirror

(2) Convex mirror

(3) Concave mirror

(4) Plano-convex mirror

23) A small coin is resting on the bottom of a beaker filled with a liquid. A ray of light from the coin travels upto the surface of the liquid and moves along its surface (see figure)



How fast is the light travelling in the liquid?

(1) 1.8×10^8 m/s

(2) 2.4×10^8 m/s

(3) 3.0×10^8 m/s

(4) 1.2×10^8 m/s

24) Two coaxial thin lenses of equal focal length of opposite nature are separated by a distance d. The combination will behave like a :-

(1) Convex lens

(2) Concave lens

(3) Plane glass plate

(4) Plane mirror

25) The light reflected by a plane mirror, may form a real image :

(1) If the ray incident on the mirror are diverging

(2) If the rays incident on the mirror are converging

(3) If the object is placed very close to the mirror

(4) Under no circumstances

26) A plane mirror is approaching you at a speed 5 cm/sec. You see your image in it. At what speed your image will approach you ?

(1) 10 cm/s

(2) 5 cm/s

(3) 20 cm/s

(4) 15 cm/s

27) All of the following statements are correct for real object, except.

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(1) The magnification produced by a convex mirror is always less than '1'

(2) A virtual 'erect' same sized image can be obtained using a plane mirror.

(3) a virtual , erect & magnified image can be formed using concave mirror.

(4) A real inverted, same sized image can be formed using a convex mirror.

28) A 2cm tall object is placed 8 cm distant infront of a mirror. In order to produce an upright image of 6 cm height one needs as :

(1) A convex mirror of radius of curvature12 cm

(2) A concave mirror of radius of curvature 12 cm

(3) A concave mirror of radius of curvature 24 cm

(4) Plane mirror of height 12 cm

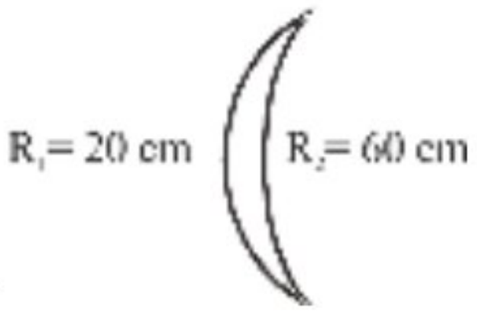
29) A beam of light is converging at a point I on a screen. A plane glass slabe whose thickness is 4 cm & refractive index is 1.5 is introduced in the path of the beam. The convergence point is shifted by.

(1) 2 cm away

(2) $\frac{4}{3}$ cm away

(3) 2 cm nearer

(4) $\frac{4}{3}$ cm nearer



30) Find focal lengths of lens made of glass ($\mu = 3/2$) and placed in air :-

(1) +60 cm

(2) –60 cm

(3) +30 cm

(4) –30 cm

31) Light travels x distance in t_1 time in air and 5x distance in t_2 time in a medium. Critical angle for this medium - air combination will be

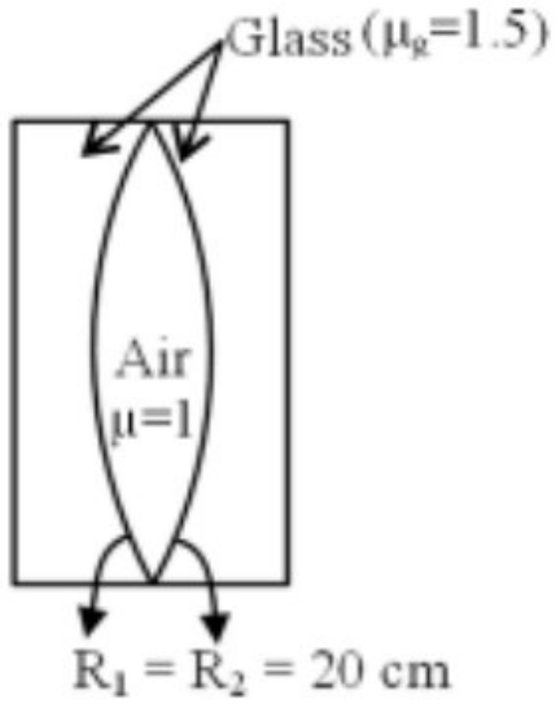
(1) $\sin^{-1}\left(\frac{t_2}{5t_1}\right)$

(2) $\sin^{-1}\left(\frac{t_1}{5t_2}\right)$

(3) $\sin^{-1}\left(\frac{5t_1}{t_2}\right)$

(4) $\sin^{-1}\left(\frac{5t_2}{t_1}\right)$

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32)
In above situation focal length will be

(1) 30 cm

(2) – 20 cm

(3) 10 cm

(4) – 10 cm

33) Angle of prism is 30° & it's refractive index is $\sqrt{2}$ & one of it's face is silvered. What should be angle of incidence for light ray to retrace it's path from silvered face ?

(1) 30°

(2) 60°

(3) 45°

(4) $\sin^{-1}\left(\frac{2}{3}\right)$

34) A ray of monochromatic light is incident on one refracting face of a prism of angle 105°. It passes through the prism and is incident on the other face at the critical angle. If the refractive index of the material of the prism is $\sqrt{2}$, the angle of incidence on the first face of the prism is

(1) 30°

(2) 45°

(3) $\sin^{-1}\sqrt{3}$

(4) $\sin^{-1}\left(\sqrt{\frac{3}{2}}\right)$

35) For a simple microscope an object is placed 5 cm away from the lens. What will be angular magnification.

(1) 5

(2) 6

(3) 4

(4) 10

36) Distance of object from first focus of equiconvex lens is 10 cm & distance of real image from

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second focus is 40 cm. If refractive index of material of lens is $\frac{3}{2}$ then find out radius of curvature of curved surface.

(1) 40 cm

(2) 10 cm

(3) 80 cm

(4) 20 cm

37) Match the column I with column II
[Where m is magnification produced by mirror]

	Column-I		**Column-II**
(A)	m = -3	(P)	Convex mirror
(B)	$m = -\frac{1}{3}$	(Q)	Concave mirror
(C)	m = + 3	(R)	Real Image
(D)	$m = +\frac{1}{2}$	(S)	Virtual Image

(1) (A) → Q,S, (B) → Q,S, (C) → Q,R, (D) → P,R

(2) (A) → Q,R, (B) → Q,R, (C) → Q,S, (D) → P,S

(3) (A) → P,S, (B) → Q,R, (C) → Q,S, (D) → P,S

(4) (A) → P,R, (B) → Q,S, (C) → P,Q,R, (D) → Q,S

38) A copper calorimeter of 150 g contains 200g of water at 40°C. A 100g aluminium piece at 95°C is dropped into it. The final temperature of system becomes 45°C. From the observed data, the specific heat of aluminium calculated is :
(Given: specific heat of copper = 400 J/kg-°C; specific heat of water = 4200 J/kg-°C)

(1) 250 J/kg-^{o}C

(2) 440 J/kg-^{o}C

(3) 600 J/kg-^{o}C

(4) 900 J/kg-^{o}C

39) A ball of thermal capacity 40 cal/°C is heated to the temperature of a furnace. It is then transferred into a copper calorimeter containing 240 g water. If water equivalent of calorimeter is 80 g and temperature of calorimeter and its contents rises from initial value of 10°C to a final value of 70°C, the temperature of furnace is :

(1) 480°C

(2) 550°C

(3) 660°C

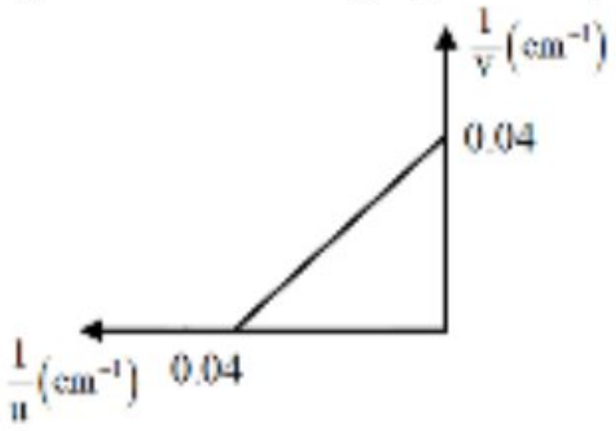
(4) 800°C

40) The focal length (f) of a thin convex lens is determined by two pin method, for no parallax. The

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numerical values of real image distances (v) are measured for set of values of object distances (u).

Figure shows the graph of $\frac{1}{v}$ (in cm^{-1}) versus $\frac{1}{u}$(in cm^{-1}). The optical power of lens is :



(1) +4D

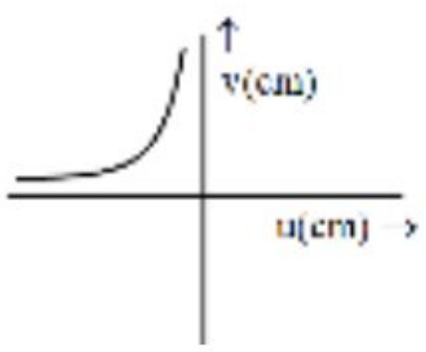
(2) +2.5D

(3) +1.25D

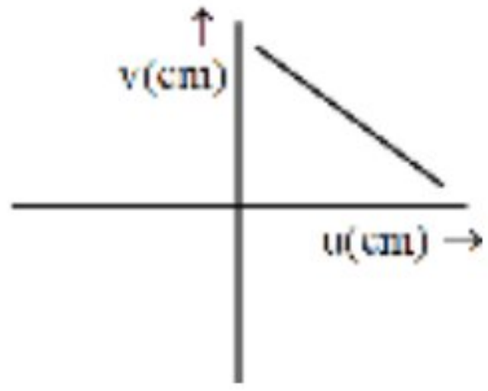
(4) +0.5D

41) A student determines the focal length of a convex lens by putting an object pin at a distance 'u' from the lens and measuring the distance 'v' of the image pin, for no parallax between two pins. The correct graph between 'u' and 'v' plotted by student will be :

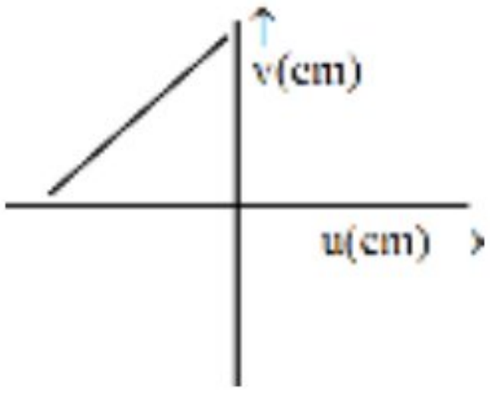
(1)



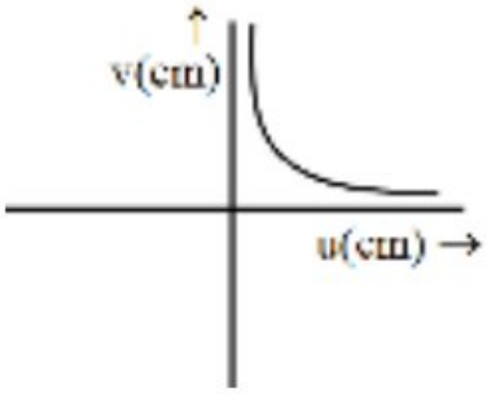
(2)



(3)

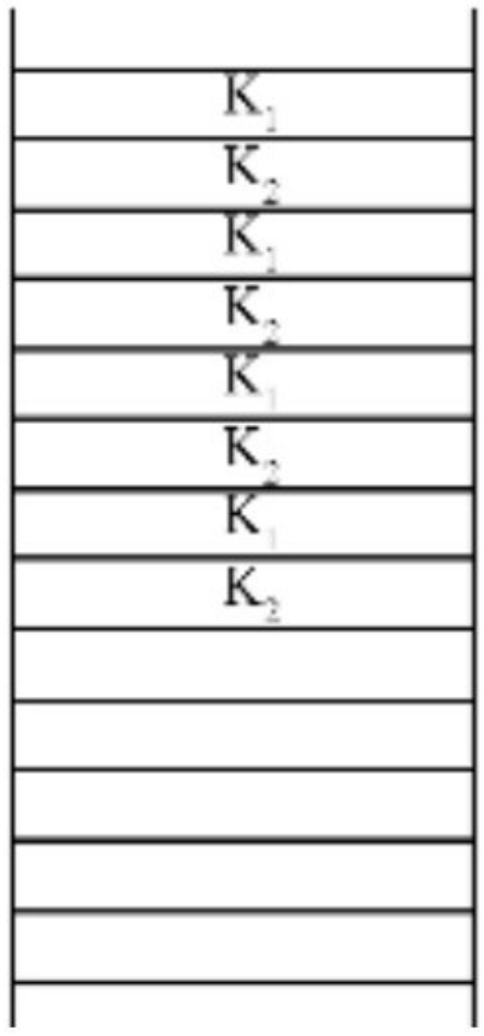


(4)



42) A wall consists of 'n' alternating blocks with length 'd' and coefficient of thermal conductivity K_1 and K_2. The cross-sectional area of the blocks are the same. The equivalent coefficient of thermal

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conductivity of the wall between left and right is :-

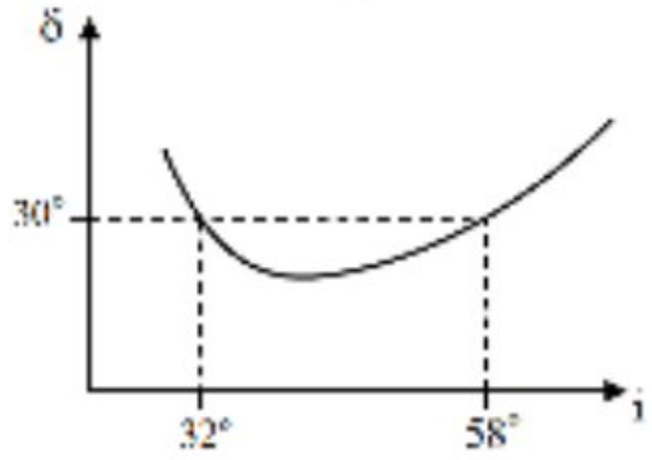
(1) $K_1 + K_2$

(2) $\frac{K_1 + K_2}{2}$

(3) $\frac{K_1 K_2}{K_1 + K_2}$

(4) $\frac{2K_1 K_2}{K_1 + K_2}$

43) Graph between angle of deviation (δ) versus angle of incidence (i), for a prism is as shown in figure. The apex angle of prism is :



(1) 30°

(2) 45°

(3) 60°

(4) 90°

44) A student determines the refractive index of a glass slab using a travelling microscope having least count of 0.001 cm with following observations :

	When microscope is focused on :	**Reading**
(A)	Mark on paper without slab	8.140 cm
(B)	Mark on paper with slab	6.925 cm
(C)	On top surface of slab with lycopodium powder	4.900 cm

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Refractive index of glass slab is :

(1) 1.4

(2) 1.5

(3) 1.6

(4) 1.65

45) In an experiment, a student put an ink mark on a paper and it is seen well focused at a distance of 45 cm by a travelling microscope. Now he put a glass slab over the mark, and he found that he has to move the microscope 6 cm away from the slab to get focused again on mark. If refractive index of glass slab is 1.5, the thickness of slab is given by -

(1) 15cm

(2) 18cm

(3) 20cm

(4) 21cm

CHEMISTRY

1) Which of the following compound is used as an adsorbent in column chromatography :-

(1) Na_2O

(2) Na_2SO_4

(3) Al_2O_3

(4) NaCl

2) In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is :-

(1) 18.20

(2) 16.76

(3) 15.76

(4) 17.36

3) The Lassaigne's extract is boiled with conc. HNO_3 while testing for the halogens. By doing so it :-

(1) Decomposes Na_2S and NaCN, if formed

(2) Helps in the precipitation of AgCl

(3) Increases the solubility product of AgCl

(4) Increases the concentration of NO_3^- ions

4) Which of following is not correctly match :-

(1) Column chromatography – Al_2O_3 is stationary phase

(2) Paper chromatography – Stationary phase is water

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(3) Thin layer chromatography – Example of partition chromatography

(4) Chromatography can be used to check purity of compounds

5) Given below are two statements :
Statement I : In 'Lassaigne's Test, when both nitrogen and sulphur are present in an organic compound, sodium thiocyanate is formed.
Statement II : If both nitrogen and sulphur are present in an organic compound, then the excess of sodium used in sodium fusion will decompose the sodium thiocyanate formed to give NaCN and Na_2S.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

6) Kjeldahl method is not applicable for the quantitative estimation of nitrogen for which compound ?

(1) Pyrole

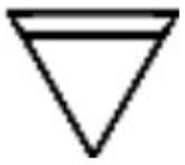
(2) Pyridine

(3) Nitrobenzene

(4) All of these

7)

Which one of the following is the most acidic ?

(1) 

(2) 

(3) 

(4) $CH_2 = CH - CH_3$

8) Identify the not feasible reaction

(1) $CH_3{-}C{\equiv}CH - \overset{\ominus}{N}H_2 \longrightarrow CH_3{-}C{\equiv}C^{\ominus} + NH_3$

(2) $CH_3{-}CH_2{-}OH + NaH \rightleftharpoons CH_3CH_2ONa + H_2$

(3) $CH_3{-}OH + NaOH \rightleftharpoons CH_3ONa + H_2O$

(4) $HC{=}CH + NaOH \rightleftharpoons HC{=}CNa + H_2O$

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9)

Which of following is electron donating group in resonance :-

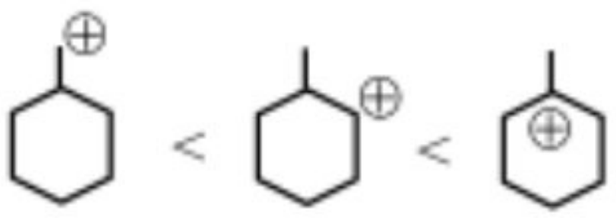
(1) $-CONHCH_3$

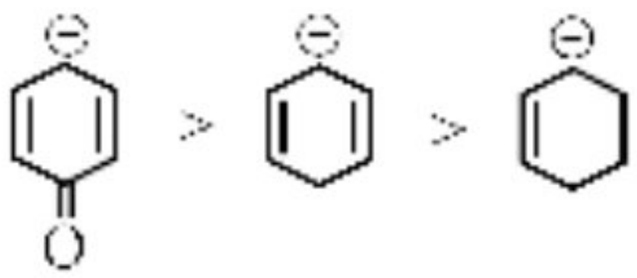
(2) –COCl

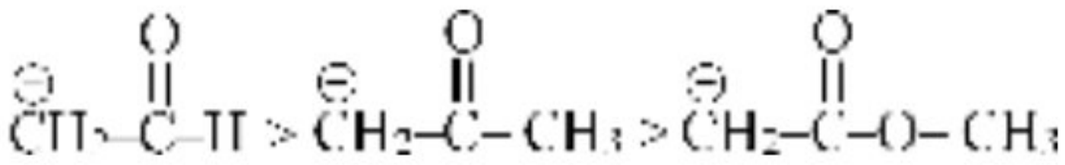
(3) –CN

(4) $-O-CH=CH_2$

10) Which of the following are correct for stability order :-

(1) 

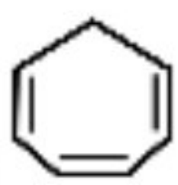
(2) 

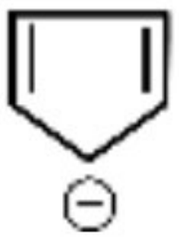
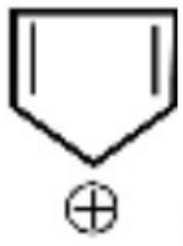
(3) 

(4) All of these

11)

Which of the following compounds are antiaromatic :-

(I)  (II)  (III) 

(IV)  (V)  (VI) 

(1) III & IV

(2) II & V

(3) I & V

(4) V & VI

12) Which of the following compound has isopropyl group:-

(1) 2,2,3,3-Tetramethyl pentane

(2) 2,2-Dimethyl pentane

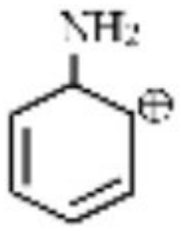
(3) 2,2,3-Trimethyl pentane

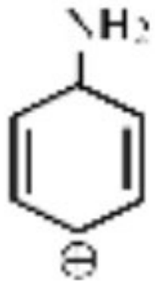
(4) 2-methyl pentane

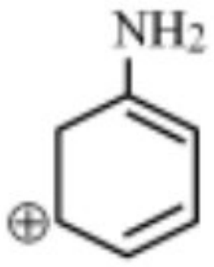
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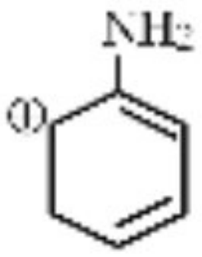
13)

The most stable carbocation for the following is :

(a) 

(b) 

(c) 

(d) 

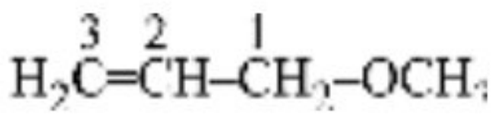
(1) c

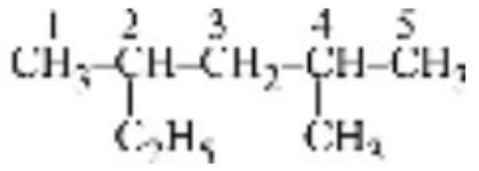
(2) d

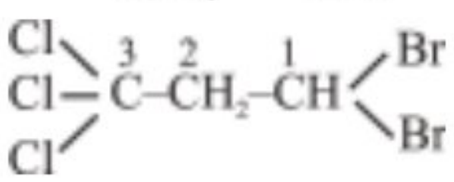
(3) b

(4) a

14) Which of the following does not represent correct numbering of C-atoms according to IUPAC system :-

(1) 

(2) 

(3) 

(4) All of these

15) Match List - I with List - II :-

	List-I (Common name)		**List-II (IUPAC name)**
(a)	Lactic acid	(i)	2-Methyl aniline
(b)	Mesityl oxide	(ii)	2-Hydroxy Propanoic acid
(c)	Acetanilide	(iii)	4-Methylpent-3-en-2-one
(d)	o-Toluidine	(iv)	N-Phenylethanamide

(1) a-(iii), b-(ii), c-(i), d-(iv)

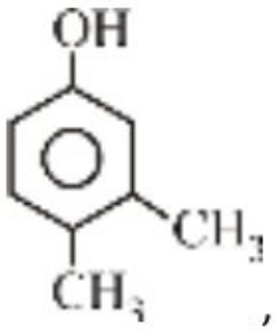
(2) a-(iv), b-(ii), c-(i), d-(iii)

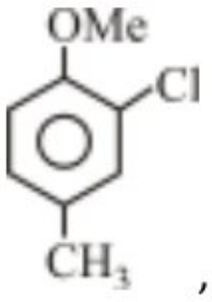
(3) a-(ii), b-(iii), c-(iv), d-(i)

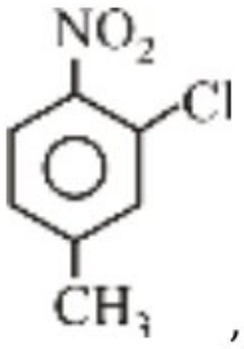
(4) a-(ii), b-(iii), c-(i), d-(iv)

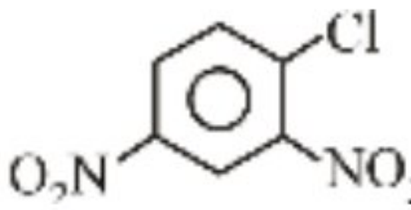
16) Which is not correctly matched :-

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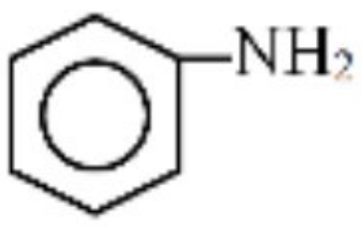
(1)  , 3,4-dimethyl phenol

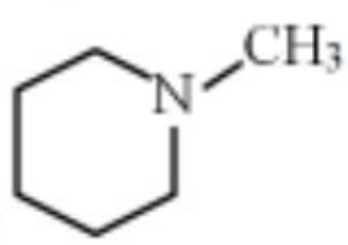
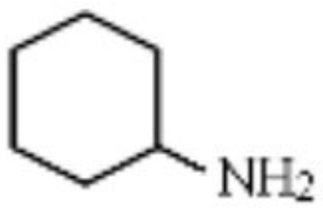
(2)  , 2-chloro-4-methyl anisole

(3)  , 2-chloro-4-methyl-1-nitrobenzene

(4)  , 2-chloro-1,5-dinitrobenzene

17) Among the following compounds, the increasing order of their basic strength is :

(I)  (II) 

(III)  (IV) 

(1) II < I < III < IV

(2) I < II < IV < III

(3) II < I < IV < III

(4) I < II < III < IV

18) Which is incorrect statement

(1) Homologous member have same functional group.

(2) Isomers have same molecular formula.

(3) Homologous members do not have same chemical properties.

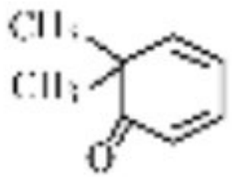
(4) Structural isomers arise due to different connectivity of atoms or group.

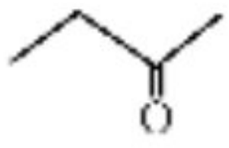
19)

Enol form for which of the following is most stable ?

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(1) 

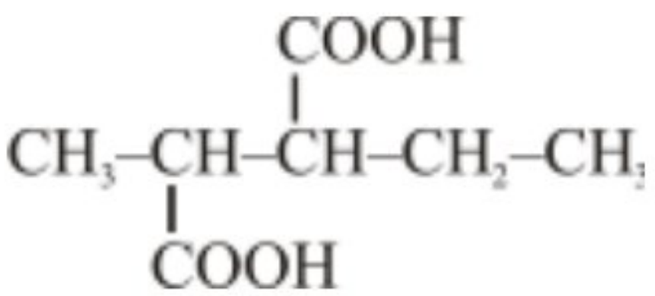
(2) 

(3) 

(4) 

20)

The IUPAC name of given compound is:-



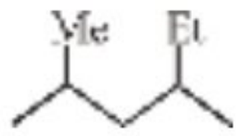
(1) 3-Methyl-2-ethylbutane-1,4-dioic acid

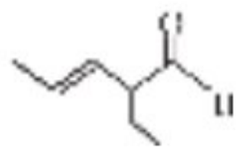
(2) 2-Ethyl-3-methylbutane-1,4-dioic acid

(3) 3-Ethyl-2-methylbutane-1,4-dioic acid

(4) Pentane-2,3-dicarboxylic acid

21) Select the incorrect match of IUPAC name with following structure :-

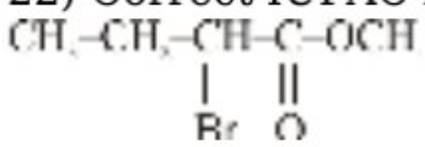
(1)  (2-Ethyl-4-methylpentane)

(2)  (2-Ethyl-pent-3-enal)

(3) CH_3COOMe (Methyl ethanoate)

(4) $HCC–CH=CH–CH_3$ (Pent-3-en-1-yne)

22) Correct IUPAC name of compound is :

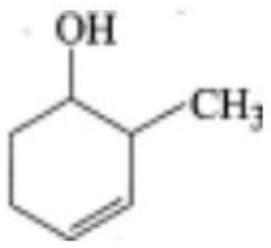


(1) 2-Bromo-1-methylbutanoate

(2) Methyl-2-bromobutanoate

(3) 1-Methyl-2-bromobutanoate

(4) Methyl-3-bromopropanoate

23) IUPAC name of the following compound is : 

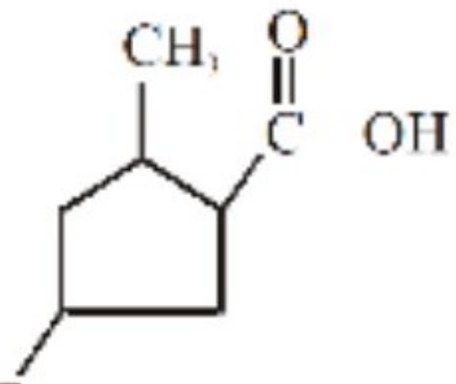
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(1) 2-methyl-3-cyclohexenol

(2) 3-methyl-1-cyclohexen-4-ol

(3) 4-hydroxy-3-methyl-1cyclohexene

(4) 2-hydroxy-1-methylcyclohexene



24) The IUPAC name of the following compound is : Br

(1) 4-Bromo-2-methylcyclopentane carboxylic acid

(2) 5-Bromo-3-methylcyclopentanoic acid

(3) 3-Bromo-5-methylcyclopentane carboxylic acid

(4) 3-Bromo-5-methylcyclopentanoic acid

25) Which one of the following compounds will not be soluble in sodium bicarbonate ?

(1) Benzene sulphonic acid

(2) Benzoic acid

(3) o-Nitro phenol

(4) 2,4,6-Trinitro phenol

26) IUPAC name of tetra-tert-butylmethane is.

(1) 3, 3-dibutyl-2,2,4,4-tetramethylpentane

(2) 3, 3-ditert-butyl-2,2,3,3-tetramethylpentane

(3) 3, 3-ditert-butyl-2, 2, 4, 4-tetramethylpentane

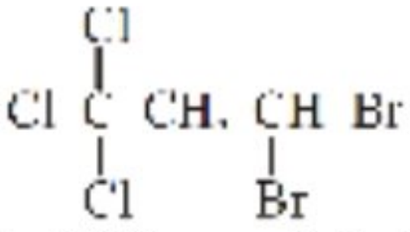
(4) 3, 3-tert-butyl-2, 2, 4, 4-tetramethylpentane

27) Correct IUPAC name is :-

(1) 

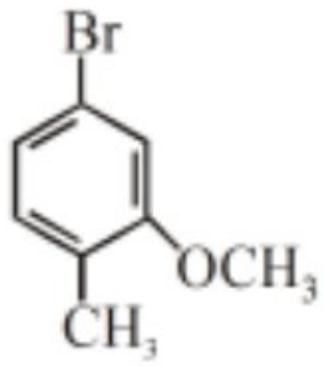
4-ethyl but-3-enoic acid

(2) 
Pent-2-en-4-yne

(3) 

1, 1-Dibromo-3, 3, 3-trichloro propane

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(4) 

4-bromo-2-methoxy-1-methyl benzene

28) ***Assertion :-*** A compound whose molecule has D-configuration will always be dextrorotatory.
Reason :- Compound having D-configuration is always rotate the plane of plane polarised light in clockwise direction.

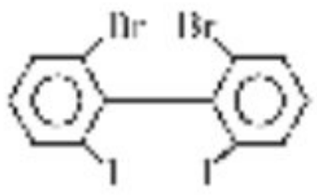
(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

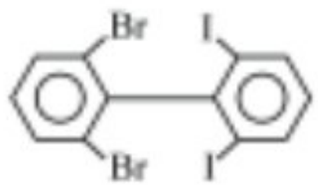
(2) **Assertion** is true but **Reason** is false.

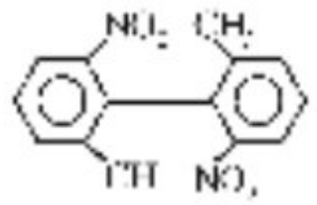
(3) **Assertion** and **Reason** both are false

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

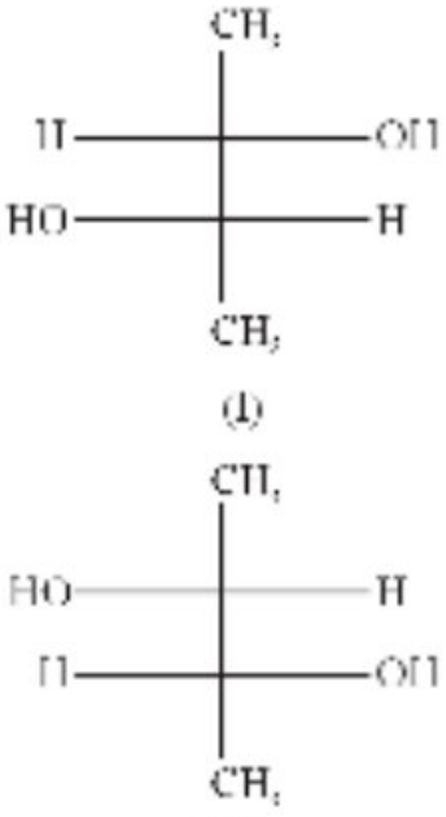
29) Which of the following biphenyl is optically active:

(1) 

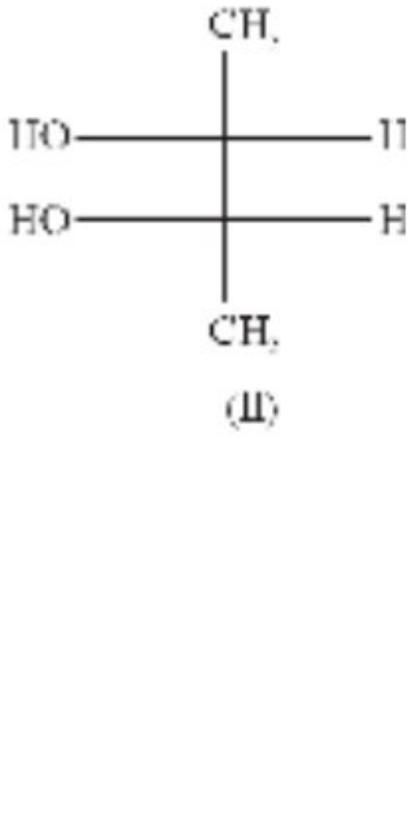
(2) 

(3) 

(4) Both (1) and (3)



(I)



(II)

(III)

30) Consider the following compounds. Choose the incorrect statements :

(1) I and III are enantiomers

(2) I and II are diastereomers

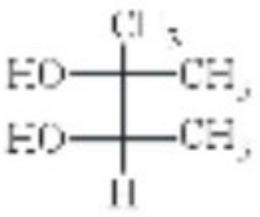
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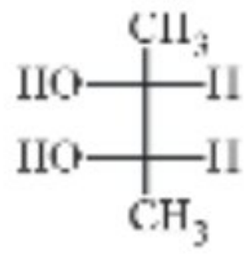
(3) II and III are diastereomers

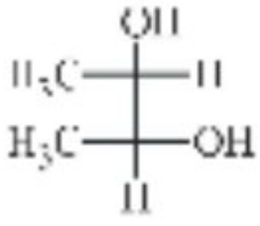
(4) I, II and III are all optically active

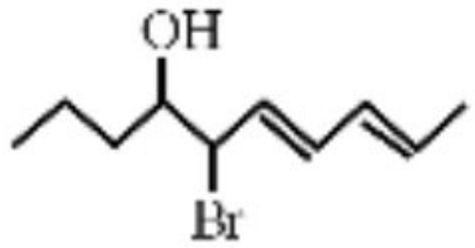
31) Which of the following compound is optically inactive:-

(1) 

(2) 

(3) 

(4) 



32) How many stereo isomer are possible in given compound.

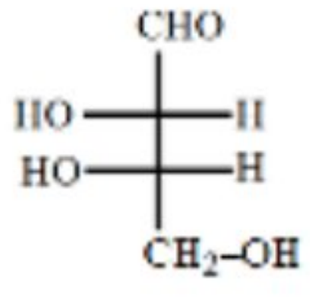
(1) 4

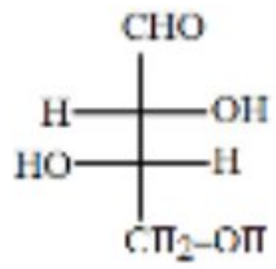
(2) 8

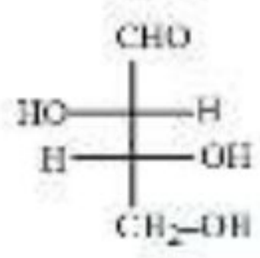
(3) 32

(4) 16

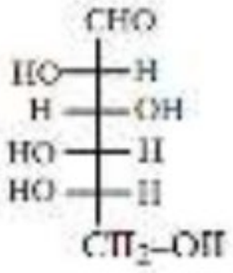
33) Which of the following is/are D form :

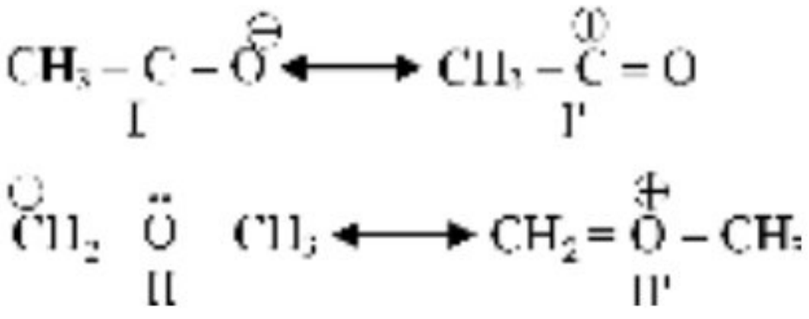
(1) 

(2) 

(3) 

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(4) 

34) 

The correct order of stability is :-

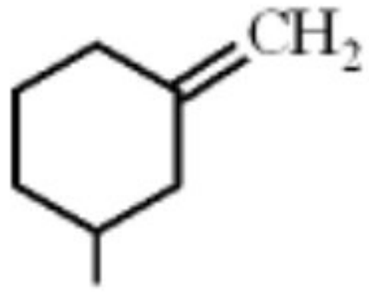
(1) I = I', II = II'

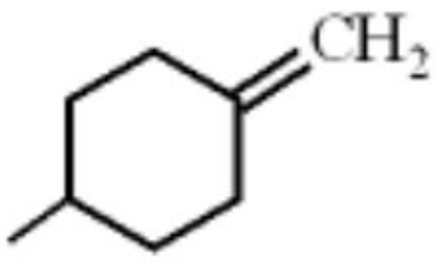
(2) I > I', II > II'

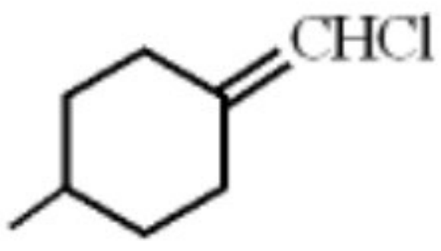
(3) I > I', II' > II

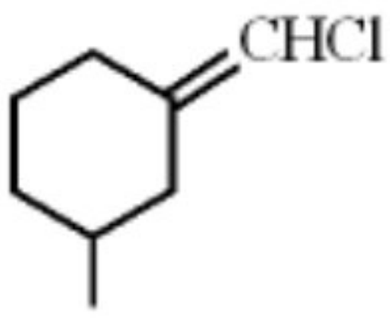
(4) I' > I, II' > II

35) The geometrical isomerism is shown by.

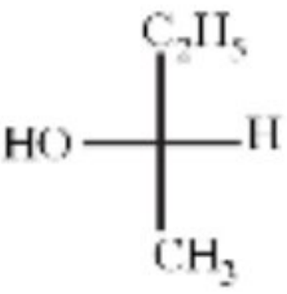
(1) 

(2) 

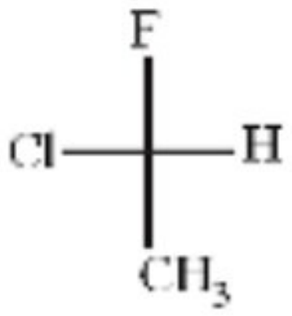
(3) 

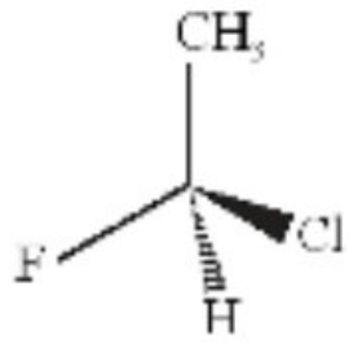
(4) 

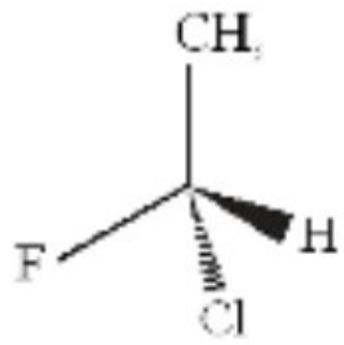
36) Which of the following have R configuration

(1) 

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(2) 

(3) 

(4) 

37) Which of the following conformor of ethane-1,2-diol is most stable

(1) Anti

(2) Gauche

(3) Partially eclipsed

(4) Fully eclipsed

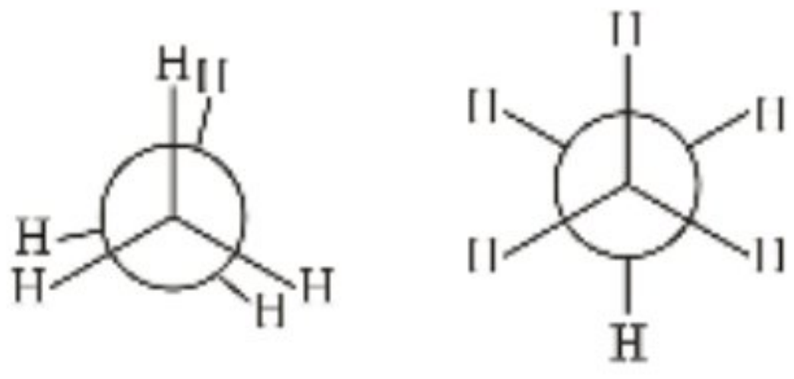
38) What should be the minimum conditions to show geometrical isomerism ?

(1) Restricted rotation about double bond or ring

(2) Groups which are responsible to show geometrical isomerism differ in their relative distance

(3) Two different groups at both restricted atoms

(4) All of these



(A) (B)

39) Incorrect statement for - (Newmann's projection of ethane)

(1) Any other intermediate conformer between A and B are called skew conformation.

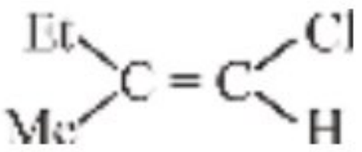
(2) A have more torsional strain than B

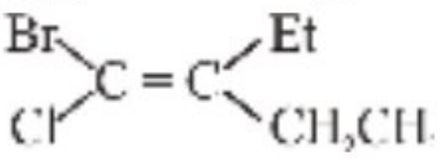
(3) Energy barrier between A and B are very small ($12.5\ kJ\ mol^{-1}$)

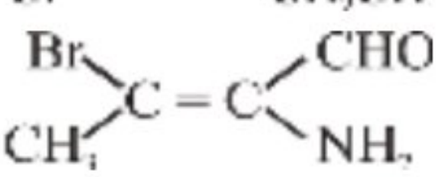
(4) Bond angle and bond length are different in A and B.

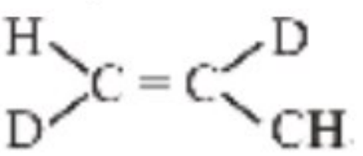
40) Which of the following is 'E' form ?

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(1) 

(2) 

(3) 

(4) 

41) How many mono-chloro product formed by isobutane (only structural) :-

(1) 1

(2) 2

(3) 3

(4) 4

42) Which of the following pairs of compounds are position isomers :-

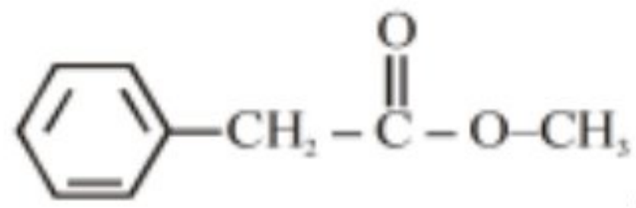
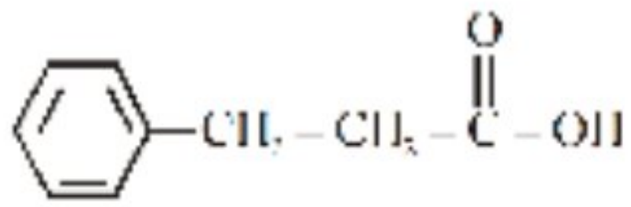
(1) Isobutyl alcohol and t-butyl alcohol

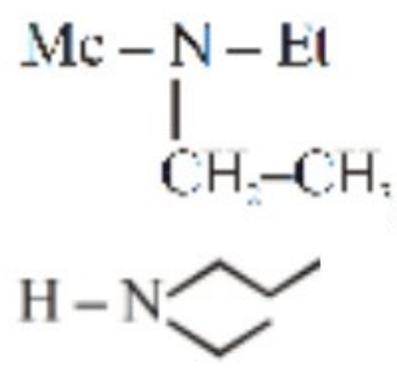
(2) Isopentyl alcohol and n-pentyl alcohol

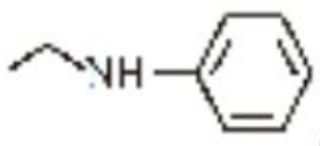
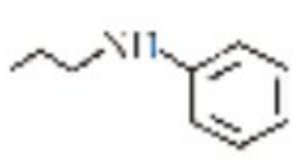
(3) Isobutyl alcohol and s-butyl alcohol

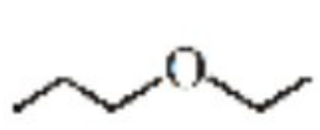
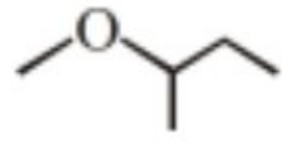
(4) α-butylene and isobutylene

43) Which of the following are metamers.

(1)  and 

(2)  and

(3)  and 

(4)  and 

44) Total number of cyclic structural isomers possible for compound having molecular formula C_6H_{12}

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:

(1) 10

(2) 11

(3) 12

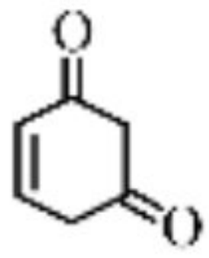
(4) 13

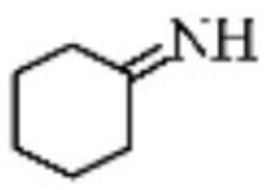
45)

Tautomerism is exhibited by :

(I)

(II)

(III) 

(IV) 

(1) I & II

(2) I, II, IV

(3) I, III, IV

(4) III only

BIOLOGY

1) Functional unit of nature is :

(1) Organism

(2) Population

(3) Community

(4) Ecosystem

2) Total sum of all ecosystems present on Earth is :

(1) Biome

(2) Global ecosystem

(3) Biosphere

(4) Both 2 and 3 given above

3) Crop fields are example of :

(1) Natural ecosystem

(2) Aquatic ecosystem

(3) Artificial ecosystem

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(4) All of the above

4) _____ is the vertical distribution of different plant species in an ecosystem.
Choose the correct option among given below and fill in the blank :

(1) Stratification

(2) Species diversity

(3) Dominance

(4) Identification

5)

Which of the following is correct unit of productivity :

(1) gm^{-2}

(2) $gm^{-2}\ y^{-1}$

(3) kcal $m^{-2}\ y^{-1}$

(4) Both 2 and 3 given above

6) _____ is the rate of formation of new organic matter by consumers.
Choose the correct option among given below and fill in the blank.

(1) Gross primary productivity

(2) Net primary productivity

(3) Secondary productivity

(4) Both 1 and 2 given above

7) The annual net primary productivity of whole biosphere is approximately _____ of organic matter.
Choose the correct option among given below and fill in the blank.

(1) 170 billion tons

(2) 55 billion tons

(3) 150 billion tons

(4) 70 billion tons

8) Nutrients which go down into soil horizon & get precipitated by leaching, are :

(1) Organic nutrients

(2) Inorganic nutrients

(3) Both organic and inorganic nutrients

(4) Nutrients never participate in leaching

9) Factor that directly affect rate of decomposition :

(1) Nature of detritus

(2) Temperature

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(3) Soil moisture

(4) All of the above

10) Of the incident solar radiation, _____ is photosynthetically active radiation.
Choose the correct option among given below and fill in the blank.

(1) 1-5 percent

(2) 2-10 percent

(3) More than 50 percent

(4) Less than 50 percent

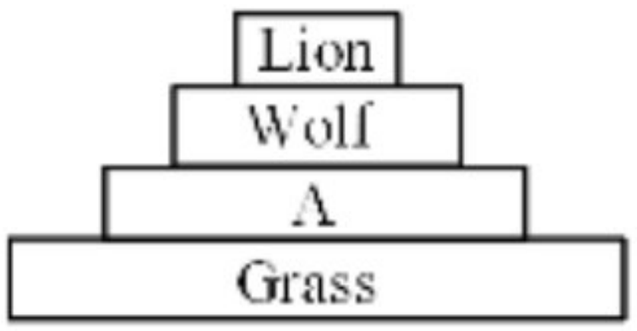
11) In an ecosystem. At first trophic level, IF value of gross primary productivity is 100 Kcal, then what will be value of Net primary productivity at first trophic level ?

(1) 100 Kcal

(2) 90 Kcal

(3) 10 Kcal

(4) 5 Kcal

12) Given below an ecological pyramid :- 

Which of the following option correctly represents A :-

(1) Phytoplankton

(2) Fox

(3) Zooplankton

(4) Cow

13) Which of the following is not a parameter of ecological pyramids ?

(1) Number

(2) Energy

(3) Fresh weight

(4) Dry weight

14) In an ecosystem, a given species may occupy _____ trophic levels.
Choose the correct option among given below and fill in the blank.

(1) One

(2) Infinite

(3) Less than one

(4) More than one

15) Given below list of some animals :

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Human, Rabbit, Sparrow, Lion
How many belong to more than one trophic levels ?

(1) One

(2) Two

(3) Three

(4) Four

16) Match the columns given below :

Column-I		Column-II	
(A)	Grass	(i)	Top consumer
(B)	Grass hopper	(ii)	Secondary consumer
(C)	Wolf	(iii)	Producer
(D)	Lion	(iv)	Primary consumer

Choose the option having all correct matching pairs :

(1) A-iii, B-iv, C-i, D-ii

(2) A-iii, B-iv, C-ii, D-i

(3) A-iv, B-iii, C-i, D-ii

(4) A-iv, B-i, C-iii, D-ii

17) _____ is the main conduit of energy transfer in terrestrial ecosystem while _____ is the main conduit of energy transfer in aquatic ecosystem.
Choose the correct option among given below and fill in the blanks respectively.

(1) GFC, DFC

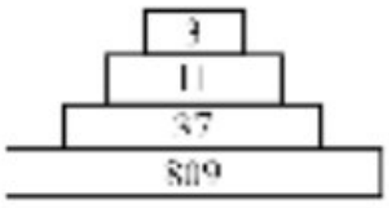
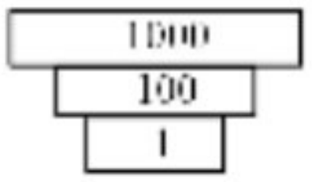
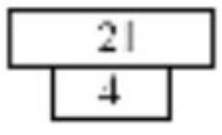
(2) GFC, GFC

(3) DFC, GFC

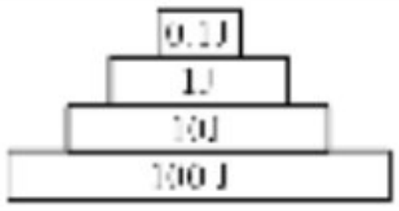
(4) DFC, DFC

18)

Given below are two Columns :-

	Column-I		Column-II
(A)	Pyramid of number of grassland	(i)	
(B)	Pyramid of biomass in lake	(ii)	
(C)	Pyramid of number in true	(iii)	

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(D)	Pyramid of energy in ocean	(iv)	

Which of the following option represent all correct matching pairs :-

(1) A-iv, B-i, C-ii, D-iii

(2) A-i, B-iii, C-ii, D-iv

(3) A-i, B-ii, C-iii, D-iv

(4) A-iv, B-ii, C-iii, D-i

19) Given below are two statements : **Statement-I :** Energy at lower trophic level is always more than that at higher trophic level.
Statement-II : Pyramid of energy can also be inverted.
In the light of above statements, choose the most appropriate answer among the options given below :

(1) Statement-I is correct but Statement-II is incorrect.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Both Statement-I and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

20) Which of the following statement is incorrect ?

(1) The base of each pyramid represents top consumers

(2) A given species may occupy more than one tropic level at same time in same ecosystem

(3) Pyramid of energy is always upright.

(4) Pyramid of biomass in sea is generally inverted.

21) Read the following statements :
(i) Only 10 percent of energy is transfered to each trophic level from lower trophic level.
(ii) Standing crop is measured in terms of biomass only.
(iii) In ecosystem, amount of energy increases at successive trophic levels.
(iv) Number of trophic levels in detritus food chain is unlimited.
How many of above statements are correct ?

(1) One

(2) Two

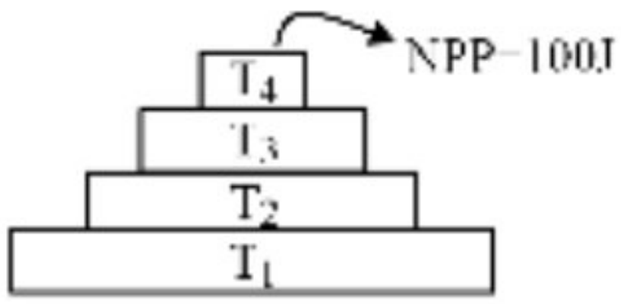
(3) Three

(4) Four

22)

Given below pyramid of energy in a grassland ecosystem :-

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Value of energy of sunlight available to producers will be ?

(1) 10^5 J

(2) 10^6 J

(3) 10^7 J

(4) 10^{10} J

23) Loss of Biodiversity may lead to all, except.

(1) Decline in plant production

(2) Increased resistance to environmental perturbance

(3) Increased variability in water use

(4) Increased variability in pest and disease cycle

24) Which of the following is not a reason that accounts for greater biodiversity of tropics ?

(1) Availability of more solar energy

(2) More niche specialization

(3) More time for species diversification

(4) Large seasonal variations in environmental factors

25) The relation between species richness and area for a wide variety of taxa on a logarithmic scale is a :

(1) Rectangular hyperbola

(2) Straight line

(3) Sigmoid curve

(4) Sine curve

26) The hot spots of biodiversity conservation are characterized by :

(1) High endemicity and high threat of extinction

(2) low endemicity and high threat of extinction

(3) high endemicity and low threat of extinction

(4) low endemicity and low threat of extinction

27) Alexander Von Humboldt described for the first time :

(1) Ecological biodiversity

(2) Law of limiting factor

(3) Species area relationships

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(4) Population growth equation

28) How many hotspots of biodiversity in the world have been identified till date by Norman Myers ?

(1) 17
(2) 34
(3) 25
(4) 43

29) The organisation which publishes the Red list of species is :

(1) ICRE
(2) IUCN
(3) UNEP
(4) WWF

30) Which one of the following pairs of geographical areas show maximum biodiversity in our country ?

(1) Sunderban and Rann of Kutch
(2) Eastern Ghats and West Bangal
(3) Eastern Himalayan and Western Ghats
(4) Kerela and Punjab

31) Which of the following statement is NOT true ?

(1) The biodiversity decrease with increasing Altitude
(2) The biodiversity decreases with increasing pollution
(3) The fishes show greatest biodiversity among vertebrates
(4) The biodiversity of bryophytes is greater than that of angiosperm.

32) An example of ex-situ conservation is ?

(1) National park
(2) Seed bank
(3) Wild life sanctuary
(4) Sacred Grove

33) Which one of the following have the highest number of species in nature ?

(1) Insects
(2) Birds
(3) Angiosperms
(4) Fungi

34) Identify the incorrect statement

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(1) Biodiversity is the term popularised by sociobiologist Edward Wilson to describe the combined diversity at all level of biological organization.

(2) IUCN stands for international union for conservation of nature and natural resources

(3) Robert may places the global species diversity in the range of 20 to 50 million.

(4) More than 25 percent of the drugs currently sold in the market are derived from the plants.

35) Amongst the vertebrates, the class with maximum number of species is ?

(1) Pisces

(2) Reptilia

(3) Aves

(4) Mammalia

36) If we consider Invertebrates the maximum global biodiversity after insects is seen in :

(1) Crustaceans

(2) Echinoderm

(3) Molluscs

(4) Annelids

37) Identify the incorrectly matched pair :-

(1) Dodo - Mauritius

(2) Quagga - Africa

(3) Thylacine - Australia

(4) Stellar's sea cow - USA

38) Which group amongst the following has the maximum percentage of species facing threat of extinction ?

(1) Amphibian

(2) Birds

(3) Mammals

(4) Gymnosperm

39) Which of the following statement is not true ?

(1) India covered 2.4% area of earth

(2) In plants maximum diversity is found in fungi

(3) In nature maximum number is of insects

(4) In plants after angiosperm maximum number is of fungi

40) Match list-I with list-II

	List-I		**List-II**
(a)	The Evil quartet	(i)	Cryopreservation

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(b)	Ex-situ conservation	(ii)	Alien species Invasion
(c)	*Lantana camara*	(iii)	Causes of Biodiversity losses
(d)	Dodo	(iv)	Extinct

Choose the option with all correct matches :

(1) a-iii, b-ii, c-i, d-iv

(2) a-iii, b-i, c-ii, d-iv

(3) a-iii, b-iv, c-ii, d-i

(4) a-iii, b-ii, c-iv, d-i

41) Match the list-I with list-II :

	List-I		**List-II**
(a)	Robert May	(i)	Species are a relationship
(b)	Alexander von Humboldt	(ii)	Long term ecosystem experiment using out door plots
(c)	Paul Ehrlich	(iii)	Global species diversity at about 7 Million
(d)	David Tilman	(iv)	Rivet popper hypothesis

Choose the correct answer from the options given below :

(1) a-ii, b-iii, c-i, d-iv

(2) a-iii, b-i, c-iv, d-ii

(3) a-i, b-iii, c-ii, d-iv

(4) a-iii, b-iv, c-ii, d-i

42) More than 200 species of cichlid fish got extinct in the lake victoria because of :-

(1) Over exploitation

(2) Alien species Invasion

(3) Habitat loss and fragmentation

(4) Co-extinction

43) **Assertion (A) :** Temperate region account for greater biological diversity as compared to tropics.
Reason (R) : Tropical environments unlike temperate ones are less seasonal, relatively more constant.

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

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(3) Assertion is true but Reason is false

(4) Assertion is false but Reason is true

44) **Assertion (A) :** A biosphere reserve is a specified area.
Reason (R) : No restiction on human activities has been imposed in biosphere reserve.

(1) Both Assertion and reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

45) Production of oxygen by forests, Is an example of :-

(1) Narrowly utilitarian

(2) Broadly Utilitarian

(3) Ethical Argument

(4) Deforestation

46) **Assertion:** Insulin can be given orally to diabetic patients.
Reason: Insulin decreases cellular glucose uptake and its utilisation.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

47) **Assertion:** Pancreas is composite gland
Reason: Pancreas acts as both exocrine and endocrine gland.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

48) **Assertion:** Graves' disease is also called as Exopthalmic goitre.
Reason: Protrusion of eye balls, enlargement of thyroid gland are the main characteristics of Grave's disease.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

49) **Assertion:** Thymus plays a major role in the development of Immune system.
Reason: Thymosine provide both cell mediated as well as humoral immunity.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

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(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

50) Diabetes insipidus occur due to an impairment affecting synthesis or release of :-

(1) Melatonin

(2) Anti-diuretic - hormone (ADH)

(3) Insulin

(4) Aldosterone

51) Match Column-I with Column-II **correctly** :-

	Column-I		**Column-II**
(A)	Protein hormone	(i)	Thyroid hormones
(B)	Steroid hormone	(ii)	Cortisol
(C)	Iodothyronines	(iii)	Epinephrine
(D)	Amino acid derivatives	(iv)	Glucagon

	A	B	C	D
(1)	iii	ii	i	iv
(2)	ii	iii	i	iv
(3)	iv	ii	i	iii
(4)	iv	iii	i	ii

(1) 1

(2) 2

(3) 3

(4) 4

52) In acromegaly especially severe disfigurement of the _____ occur.

(1) Face

(2) Chest

(3) Hand

(4) Leg

53) Which of the following hormones is synthesized by hypothalamus ?

(1) Prolactin, Oxytocin

(2) Oxytocin, ADH

(3) FSH, GnRH

(4) LH, FSH

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54) **Assertion :** The secretion of PTH is regulated by the circulating levels of calcium ions.
Reason : Parathyroid hormone (PTH) decreases the Ca^{2+} levels in the blood.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is true but reason is false.
(4) Both Assertion & Reason are False.

55) Which hormone stimulates the pituitary for synthesis and secretion of gonadotrophins ?

(1) Estrogen
(2) GnRH
(3) TSH
(4) MSH

56) Which of the following is correct ?

(1) Thyroxin hypersecretion is associated with hypoglycemia.
(2) Insulin acts on pancreatic cells and adipocytes.
(3) Aldosteron hypersecretion is associated with myosthenia gravis.
(4) Glucocorticoids stimulate gluconeogenesis.

57) Select the incorrect statement from following options :

(1) Hormones are non-nutrient chemicals
(2) Endocrine glands are with ducts & are hence called ducted gland
(3) Hormones are produced in trace amount
(4) Hormone act as Intercellular messanger.

58) **Statement-I :-** Progestational hormone helps to prepare the uterus for implantation.
Statement-II :- Progesterone supports preganancy.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and Statement-II are incorrect
(3) Statement-I is incorrect and statement-II is correct
(4) Statement-I is correct and statement-II is incorrect

59) **Assertion (A) :-** Stunted growth resulting in pituitary dwarfism is due to low secretion of somatotrophin hormone.
Reason (R) :- Somatotrophin hormone stimulates the body growth and elongation of long bones.

(1) Both **(A)** and **(R)** are correct.
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are incorrect.

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60) **Statement-I :-** JG cells of the kidney produce erythropoietin hormone.
Statement-II :- ANF is introduced in circulation when blood pressure increases.

(1) Both statement I & II are correct

(2) Both Statement I & II are incorrect

(3) Statement I is correct & II is incorrect

(4) Statement I is incorrect & II is correct

61) **Statement-I :-** Pituitary gland is located in a cartilagenous cavity called sella tursica.
Statement-II :- In humans pars intermedia is a seperate portion of pituitary.

(1) Both statement I & II are correct

(2) Both statement I & II are incorrect

(3) Statement I is correct & II is incorrect

(4) Statement I is incorrect & II is correct

62) Match the column and select correct option :

	Column-I (Hormone)		**Column-II (Target gland)**
(i)	Thyrotrophin (TSH)	(a)	Pituitary gland
(ii)	Melanotrophin (MSH)	(b)	Gonades
(iii)	Gonadotrophin (LH, FSH)	(c)	Adrenal cortex
(iv)	Corticotrophin (ACTH)	(d)	Melanocyte
(v)	Hypothalamic hormone	(e)	Thyroid

Option :

(1) i-b, ii-c, iii-a, iv-e, v-f

(2) i-e, ii-d, iii-a, iv-c, v-b

(3) i-a, ii-b, iii-d, iv-e, v-c

(4) i-e, ii-d, iii-b, iv-c, v-a

63) Which of the following pair of hormones is not antagonistic ?

(1) Insulin and glucagon

(2) FSH and LH

(3) Calcitonin and PTH

(4) MSH and Melatonin

64) Which of the following is related to hypersecretion of parathormone ?

(1) Bone resorption

(2) Renal stone

(3) Hypercalcemia

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(4) All of the above

65) Which of the following hormone decreases the blood glucose level ?

(1) Aldosterone
(2) ADH
(3) ANF
(4) Insulin

66) Which of the following hormone is secreted by anterior pituitary?

(1) Oxytocin
(2) Growth hormone
(3) Vasopressin
(4) ADH

67) Which of the following hormone is related with defence capability?

(1) FSH
(2) vasopressin
(3) Melatonin
(4) Insulin

68) **Assertion (A) :** FSH acts upon ovarian follicles in female and Leydig cells in male.
Reason (R) : Growing ovarian follicles secrete estrogen in female while interstitial cells secrete androgen in male human being.

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

69)

Match List I with List II.

List I		**List II**	
A	Exophthalmic goiter	I	Excess secretion of cortisol, moon face & hyperglycemia.
B	Acromegaly	II	Hypo-secretion of thyroid hormone and stunted growth.
C	Cushing's syndrome	III	Hyper secretion of thyroid hormone & protruding eye balls.
D	Cretinism	IV	Excessive secretion of growth hormone.

Choose the correct answer from the options given below:

(1) A-I, B-III, C-II, D-IV
(2) A-IV, B-II, C-I, D-III
(3) A-III, B-IV, C-II, D-I

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(4) A-III, B-IV, C-I, D-II

70) **Statement-I :-** Hormones are non nutrient chemicals.
Statement-II :- Hormones act as intercellular messengers.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

71) Read the following statement :
(A) Glucagon is secreted by β-cell
(B) Atrial wall of heart produce eythropoietin
(C) GIP inhibits gastric secretion and motility
(D) CCK acts on both pancreas and gall bladder
Choose the correct option :

(1) A and B

(2) B and C

(3) C and D

(4) B and D

72) Read following terminology of Column-I & II and identify their correct match-

	Column-I		**Column-II**
A.	Adrenalin	(I)	Reabsorption of Na^+ and water
B.	Glucocorticoids	(II)	Growth of axial hair pubic hair and facial hair during puberty
C.	Androgenic steroids	(III)	Mainly act on renal tubules.
D.	Aldosterone	(IV)	Lipolysis and proteolysis
		(V)	Increase heart rate
		(VI)	Ca^{+2} reabsorption

(1) (A-IV), (B-V), (C-II), (D-III)

(2) (A-VI), (B-IV), (C-III), (D-I)

(3) (A-V), (B-IV), (C-II), (D-I, III)

(4) (A-III), (B-IV), (C-II), (D-IV)

73) Hormone that interact with_____ normally do not enters target cell but generates_____messenger

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(1) membrane bound receptor, secondary
(2) membrane bound receptor, primary
(3) nuclear receptor, secondary
(4) nuclear receptor, primary

74) Which hormone produces anti-inflammatory reactions and suppresses the immune response.

(1) Thyroxine
(2) Cortisol
(3) Parathyroid hormone
(4) Thymosine

75) _____acts mainly on the liver cells and stimulates glycogenolysis resulting in hyperglycemia.

(1) Thymosin
(2) Insulin
(3) Glucagon
(4) Somatostatin

76) Hormone 'X' stimulate the process of bone resorption and increase 'Y' absorption from digested food.

(1) X → Thymosin; Y→ Potassium
(2) X → Adrenaline; Y → Vitamin B_{12}
(3) X → Parathyroid; Y → Calcium
(4) X → Thyroxine ; Y → Iodine

77) Grave's disease is caused by-

(1) Hyperthyroidism
(2) Hypothyroidism
(3) Excess secretion of growth hormone
(4) Less secretion of growth hormone.

78) Which of the following is false ?

(1) Prolactin regulates the growth of mammary glands.
(2) Posterior pituitary is under direct neural regulation of hypothalamus.
(3) MSH acts on osteocytes and regulates pigmentation of skin.
(4) Pineal gland secretes a hormone called melatonin

79) Which of the following pair of Hormone interacts with intracellular receptor?

(1) FSH & LH
(2) Prolactin & oxytocin
(3) Thyroxin & Aldosterone

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(4) Adrenaline & nor adrenaline

80) The location of hypothalamus is :-

(1) Dorsal part of midbrain
(2) Ventral part of hindbrain
(3) Basal part of diencephalon
(4) Dorsal part of diencephalon

81)

Which of the following does not transport axonally to pars nervosa-

(1) Prolactin
(2) ADH
(3) Oxytocin
(4) Vasopressin

82)

Epinephrine regulates all the following functions except-

(1) Piloerection
(2) Glycogenolysis
(3) Glycogenesis
(4) Proteolysis and lipolysis

83)

Which of the following hormone is secreted by the stromal tissue of thyroid gland ?

(1) Thyroxine
(2) Triiodothyronine
(3) Both 1 & 2
(4) None of these

84)

Which gland is present in sella tursica of cranium.

(1) Pineal gland
(2) Adrenal gland
(3) Pituitary gland
(4) Hypothalamus

85)

Which gland undergoes atrophy in response to advancing age

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(1) Pituitary gland
(2) Thyroid gland
(3) Thymus gland
(4) Adrenal gland

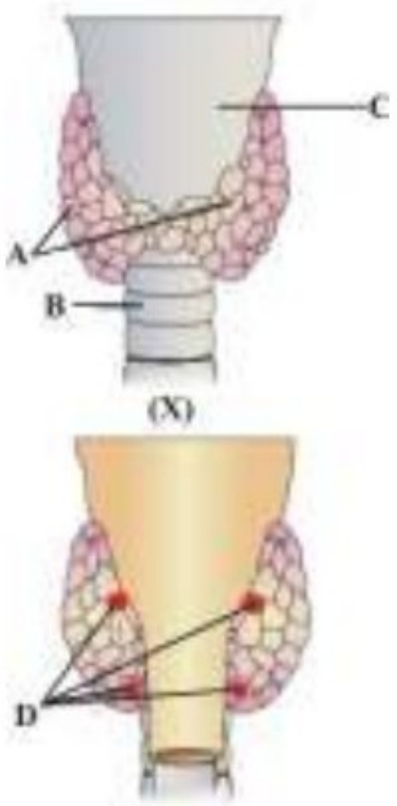
86)

What may be the cause fo malfunctioning of thyroid gland?

(1) Iron deficiency
(2) Vitamin C deficiency
(3) Iodine deficiency
(4) Calcium deficiency

87) Glucocorticoids stimulate :

(1) Release of insulin
(2) Increase GFR
(3) Lipogenesis
(4) Enhancing glucose formation



88) Select the correct option :-

(1) A–Ventro lateral position of thyroid gland,
D–Dorsally located parathyroid

(2) A–Dorso lateral position of thyroid,
D–Ventrally located parathyroid

(3) A–Dorsally located thyroid,
B–Dorsally located trachea

(4) A–Dorsally located parathyroid,
C–Epiglottis

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89) Find out correct option for point 'A' in following diagram?



(1) It’s hormone regulates dark pigmentation

(2) It’s hormone regulates calcium level in bone

(3) It’s hormone regulates survival during emergency

(4) It’s hormone inhibits androgen secretion

90) Corpus luteum secretes a hormone called :-

(1) Prolactin

(2) Progesterone

(3) Aldosterone

(4) Testosterone

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**4**	**4**	**4**	**4**	**4**	**2**	**1**	**2**	**3**	**4**	**1**	**1**	**1**	**3**	**2**	**3**	**1**	**2**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**2**	**1**	**1**	**2**	**1**	**4**	**3**	**2**	**1**	**3**	**2**	**3**	**4**	**1**	**4**	**2**	**4**	**2**	**1**
Q.	41	42	43	44	45															
A.	**1**	**2**	**3**	**3**	**2**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**3**	**2**	**1**	**3**	**1**	**4**	**2**	**4**	**4**	**4**	**4**	**4**	**1**	**4**	**3**	**4**	**3**	**3**	**1**	**2**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**1**	**2**	**1**	**1**	**3**	**3**	**4**	**3**	**4**	**4**	**3**	**4**	**3**	**3**	**4**	**3**	**2**	**4**	**4**	**3**
Q.	86	87	88	89	90															
A.	**2**	**1**	**4**	**3**	**3**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**4**	**4**	**3**	**1**	**4**	**3**	**1**	**3**	**1**	**4**	**3**	**4**	**3**	**4**	**2**	**2**	**3**	**2**	**1**	**1**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**1**	**3**	**2**	**4**	**2**	**1**	**3**	**2**	**2**	**3**	**4**	**2**	**1**	**3**	**1**	**3**	**4**	**1**	**2**	**2**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**2**	**2**	**4**	**3**	**2**	**4**	**1**	**1**	**1**	**2**	**3**	**1**	**2**	**3**	**2**	**4**	**2**	**1**	**1**	**1**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**2**	**4**	**2**	**4**	**4**	**2**	**3**	**4**	**4**	**4**	**3**	**3**	**1**	**2**	**3**	**3**	**1**	**3**	**3**	**3**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**1**	**3**	**4**	**3**	**3**	**3**	**4**	**1**	**3**	**2**										

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SOLUTIONS

PHYSICS

1) $\mu = \frac{\sin\frac{(A+\delta_{min})}{2}}{\sin\frac{A}{2}} = \sin\frac{\left(\frac{60^o+37^o}{2}\right)}{\sin 30^o}$

$= \frac{\sin(48.5^o)}{\sin 30^o}$

$= \frac{3}{4} \times \frac{2}{1} = 1.5$

2) $\frac{\theta}{A_t} = \frac{K(\theta_1 - \theta_2)}{d} = \text{const.}$

$\Rightarrow \theta_1 - \theta_2 = 20°C$

$\therefore K_A\left(\frac{\theta_1 - \theta}{d}\right) = K_B\left(\frac{\theta - \theta_2}{d}\right)$ $3K_B(\theta_1 - \theta) = K_B(\theta - \theta_1 + 20)$

$4(\theta_1 - \theta) = 20$

$\theta_1 - \theta = 5°C$

3) $C = \frac{\Delta Q}{\mu T}$

It is Q property of material and varies with temperature.

4) $\lambda_{red} > \lambda_{blue}$

From wein's law

$\lambda_m \propto \frac{1}{T}$

$T_{Red} < T_{Blue}$

5) $E = N \times \frac{3}{2}KT = q.V$

$1 \times \frac{3}{2}KT = 1.6 \times 10^{-19} \times 1$

$\Rightarrow T = 7.7 \times 10^3\ K$

6) Real exp. of liquid in A = Real exp. of liquid in B.

$\Rightarrow y_{r_1} = y_{r_2}$

$\Rightarrow y_1 + 3\alpha_A = y_2 + 3\alpha_B$

$\Rightarrow \alpha_B = \frac{y_1 - y_2}{3} + \alpha$

7) $VT^2 = \text{constant}$

$\frac{\Delta V}{V} + \frac{2\Delta T}{T} = 0$

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$$\frac{\Delta V}{V} = \frac{-2\Delta T}{T}$$

$$\frac{\Delta V}{V\Delta T} = \frac{-2}{T} = \gamma$$

$$\boxed{\gamma = \frac{-2}{T}}$$

8) $R_1R_4 = R_2R_3$

$$\frac{t}{K_1A}\frac{t}{K_4A} = \frac{t}{K_2A}\frac{t}{K_3A}$$

$K_1K_4 = K_2K_3$

9) $$\frac{\Delta w}{\Delta Q} = \frac{\mu R\, dT}{\mu C_P dT} = \frac{R}{C_P} = \frac{C_P - C_V}{C_P}$$

$$\frac{\Delta w}{\Delta Q} = 1 - \frac{1}{\gamma}$$

10)

$$\text{Water eq.} = \frac{m_{Al}.s_{Al.}}{s_{water}}$$

$$= \frac{120 \times 0.2}{1}$$

$= 24$ gm

11)

According to Prevost theory of heat exchange, all bodies radiate thermal radiations at all temperatures. The amount of thermal radiations radiated per unit time depends on the nature of the emitting surface, its area and its temperature. The rate is faster at higher temperatures.

12) $$\frac{\rho_1}{\rho_2} = \frac{3}{4}; \frac{S_1}{S_2} = \frac{4}{3}$$

$$\frac{H_{C_1}}{H_{C_2}} = \frac{m_1s_1}{m_2s_2} = \frac{V\rho_1S_1}{V\rho_2S_2} = \frac{3}{4} \times \frac{4}{3} = 1:1$$

13) $$V_{rms} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^2 + V_4^2}{N}}$$

$$= \sqrt{\frac{4V^2 + 100V^2 + 121V^2 + 75V^2}{4}}$$

$$= V\sqrt{75}$$

$$= 5\sqrt{3}V$$

14) $$\frac{\Delta Q}{W} = \frac{\mu C_P \Delta T}{\mu R \Delta T} = \frac{5/2\ R}{R} = 2.5$$

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15) W_1 = +ve, W_2 = 0, W_3 = –ve Flot ⇒ Q = W + ΔU
ΔU same for all processes
So, $Q_1 > Q_2 > Q_3$

16) Let required time = t_0
Rate of heat absorption = R
Heat absorbed in time t_0 = Rt_0
Rt_0 = Heat to make solid + Heat required to melt solid
$Rt_0 = mC\Delta T + mL$

$$t^0 = \frac{m(C\Delta T + L)}{R}$$

17) $P \propto T^4$

$$\frac{P_1}{P_2} = \left(\frac{T_1}{T_2}\right)^4 = \left(\frac{\lambda_2}{\lambda_1}\right)^4$$

$$\frac{\lambda_2}{\lambda_0} = \left(\frac{81}{256}\right)^{1/4} \Rightarrow \lambda_2 = \frac{3}{4}\lambda_0$$

$$\lambda_2 - \lambda_1 = \frac{3}{4}\lambda_0 - \lambda_0$$

$$= \frac{-\lambda_0}{4}$$

18) For earth at equilibrium
Heat energy absorb per unit time = Heat energy radiate per unit time

$$\frac{\sigma \times 4\pi R^2 . T_s^4}{4\pi d^2} \times \pi r^2 = \sigma \times 4\pi r^2 \times T_e^4$$

$$\frac{\sigma R^2 T_s^4}{d^2} = 4T_e^4$$

$$T_s = \sqrt{\frac{2d}{R}} T_e$$

19) $dQ = nC\,dT$

$$Q = \int dQ$$

$$Q = \int nC dT$$

$$= \int 2 \times 4T \times 10^{-3} dT$$

$$= 8 \times 10^{-3} \int T dT$$

$$= \frac{8 \times 10^{-3}}{2} \left[T^2\right]_{100}^{200}$$

$$= 4 \times 10^{-3} [40000 - 10000]$$

$$= \frac{4}{1000} \times 30000$$

$$= 120\ J$$

20)

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$$\text{Area} = \pi r^2 = \pi\left[\frac{h}{\sqrt{\mu^2-1}}\right]^2$$

$$=3.14\times\left[\frac{20}{\sqrt{(1.5)^2-1}}\right]^2$$

21) **Question Explanation:** We are given the absolute refractive indices of glass and water and need to find the ratio of velocities of light in glass and water.

Concept: Snell's law (refractive index and velocity of light)

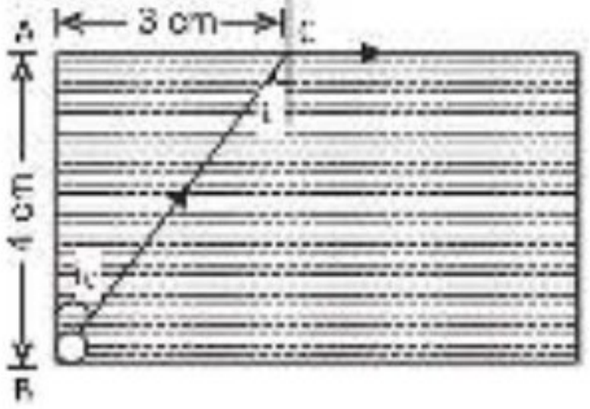
Formula: $\frac{v_{glass}}{v_{water}}=\frac{\mu_{water}}{\mu_{glass}}$

$$\frac{v_{glass}}{v_{water}}=\frac{1.33}{1.5}\approx\frac{8}{9}$$

22) $m=+\frac{1}{5}$

Erected and diminished image is formed only by convex mirror.

23)

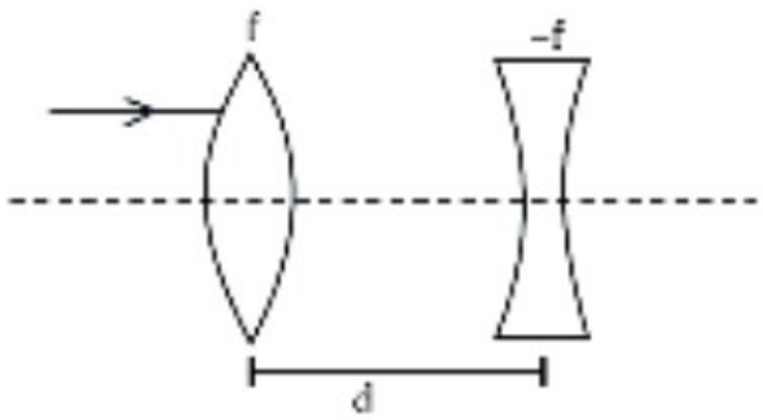


Formula : $\sin i_c=\frac{1}{\mu}=\frac{V_m}{V_a}$

Solution : by figure $BC=\sqrt{AB^2+AC^2}=\sqrt{3^2+4^2}=5cm$

$$\Rightarrow \sin i_C=\frac{AC}{BC}=\frac{3}{5}$$

$$\Rightarrow \frac{V_m}{V_a}=\frac{3}{5}\Rightarrow V_m=\frac{3\times3\times10^8}{5}=1.8\times10^8 m/s$$



24)

Formula : Net focal length of combination of two lenses separated at distance 'd'

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$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

Solution :

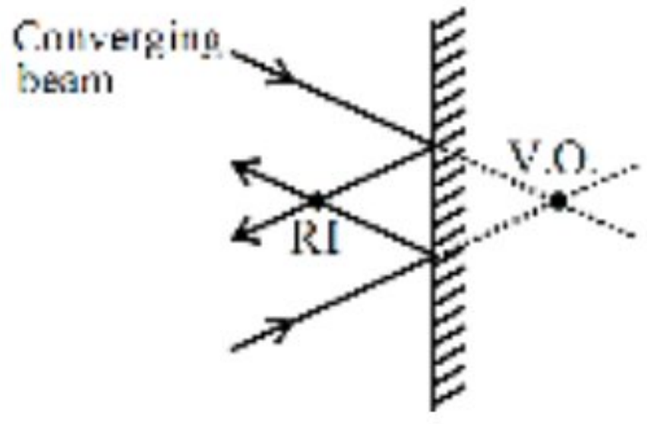
Given $f_1 = f$, $f_2 = -f$

$$\Rightarrow \frac{1}{F} = \frac{1}{f} - \frac{1}{f} - \frac{d}{f \times (-f)}$$

$$\Rightarrow \boxed{\frac{1}{F} = \frac{d}{f^2}} \text{ Power } \boxed{P = \frac{f^2}{d}} \text{ +ve value}$$

So it will behave like convex lens.

25)



26) $V_I = 2V_M$

$V_I = 2\ (5) = 10$ cm/s

27)

Convex mirror can form only virtual erected and diminished images for real objects. It can not form real inverted and same sized images.

28) $m = +\frac{6}{2} = +3$

$$\frac{f}{f-u} = +3$$

$$\frac{f}{f+8} = 3$$

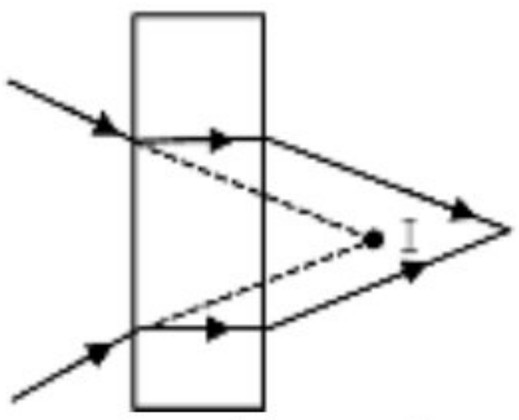
$f = 24 + 3f$

$-2f = 24$

$f = -12$ cm

$R = 2f = -24$ cm

29)

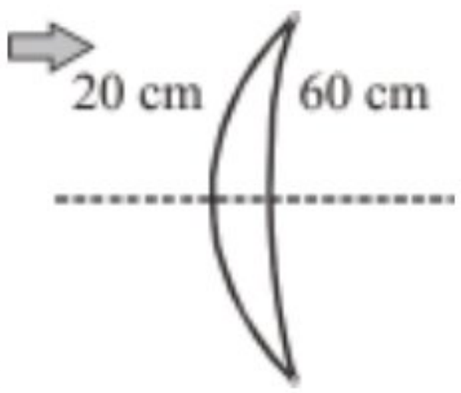


Shift $= t\left(1-\frac{1}{\mu}\right) = 4\left(1-\frac{2}{3}\right)$

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$= \frac{4}{3}$cm away

30)



$R_1 = 20$ cm, $R_2 = 60$ cm

$$\frac{1}{f} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{20} - \frac{1}{60}\right) = \left(\frac{1}{2}\right)\left(\frac{2}{60}\right)$$

$\Rightarrow f = 60$ cm

31) $$\sin\theta_c = \frac{\mu_{air}}{\mu_{med}} = \frac{V_{med}}{V_{air}}$$

$$\sin\theta_c = \frac{5x/t_2}{x/t_1} = \frac{5x}{t_2} \times \frac{t_1}{x}$$

$$\sin\theta_c = \frac{5t_1}{t_2}$$

$$\theta_c = \sin^{-1}\left(\frac{5t_1}{t_2}\right)$$

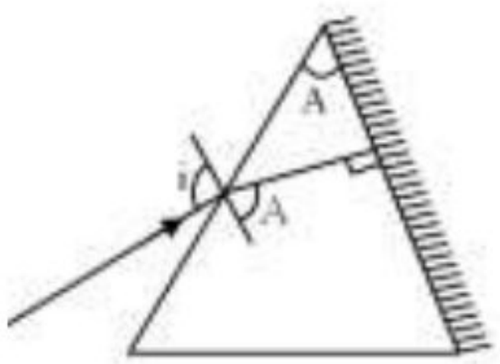
32) $$\frac{1}{f} = \frac{\mu_2 - \mu_1}{R} + \frac{\mu_1 - \mu_2}{-R}$$

$$\frac{1}{f} = \frac{1 - 1.5}{20} + \frac{1.5 - 1}{-20}$$

$$\frac{1}{f} = -\frac{1}{40} - \frac{1}{40} = -\frac{1}{20}$$

$= -20$ cm

33)



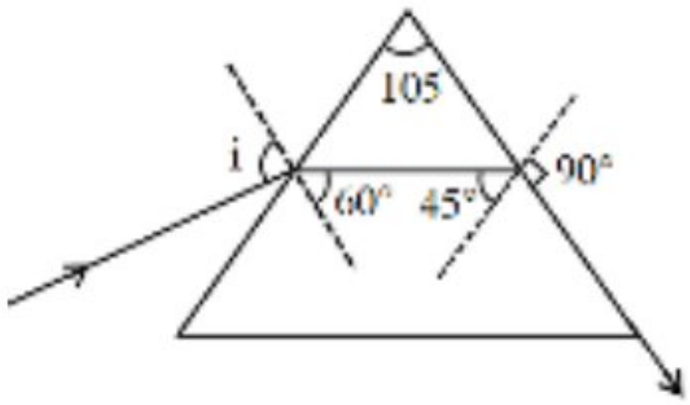
$\sin i = \mu \sin A$

$\sin i = \sqrt{2} \sin 30°$

$$\sin i = \frac{1}{\sqrt{2}}$$

$i = 45°$

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34) $\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$

$i_c = 45°$

$A = r_1 + r_2$

$105 = r_1 + 45$

$r_1 = 60$

$\sqrt{2} = \frac{\sin i}{\sin 60}$

$i = \sin^{-1}\left(\sqrt{\frac{3}{2}}\right)$

35) u = 5 cm $M.P = \frac{D}{u} = \frac{25}{5}$

MP = 5

36) x_1 = 10 cm, x_2 = 40 cm

$f = \sqrt{x_1 x_2} = \sqrt{400}$

f = 20 cm

$\frac{R}{2(1.5-1)} = 20$

R = 20 cm

37)

(A) m = –3 (inverted, real magnified image) formed by concave mirror.

(B) $m = -\frac{1}{3}$ (inverted, real diminished image) formed by concave mirror.

(C) m = + 3 (erected, virtual magnified image) formed by concave mirror.

(D) $m = +\frac{1}{2}$ (erected, virtual diminished image) formed by convex mirror.

38) Let the specific heat of aluminium be S_{Al}

From principle of calorimetry ⇒ Heat lost = Heat gained.

$(0.100)\,(S_{Al})\,(95–45) = (0.2 \times 4200 + 0.15 \times 400) \times (45–40)$

$S_{Al} = \frac{(840+60)\,(5)}{(0.100)\,(50)} = 900\,J/kg\,K$

39) Let the furnace temperature is T

Thermal capacity of ball = ms = 40cal/°C;

Final temp = 70°C

According to principle of calorimetry,

Heat lost = Heat gained

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$40(T - 70) = (240 + 80)(70 - 10)$

$(T - 70) = 480 \Rightarrow T = 550^{o}C$

40) In two pin method, the image formed will be real and inverted for no parallax.

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{1}{u} + \frac{1}{f}$$

on comparing equation(1) with y = mx + c,

$m = +1, \frac{1}{f}$ = intercept

Here, intercept $= \frac{1}{f} = 0.04\ (cm^{-1})$

$$\Rightarrow f = \frac{1}{0.04} = 25cm$$

$$P = \frac{100}{f} = \frac{100}{25} = +4D$$

41) From lens equation

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow v = \frac{uf}{(u+f)} \Rightarrow v = \frac{f}{\left(1 + \frac{f}{u}\right)}$$

(i) u = ∞; v = f

(ii) u = -f; v = ∞

42) It's a parallel combination

$$R_1 = \frac{d}{K_1 A} \quad R_2 = \frac{d}{K_2 A}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + ...\text{upto } n^{th}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{n}{2R_1} + \frac{n}{2R_2}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{n}{2}\frac{R_1 + R_2}{R_1 R_2}$$

$$\Rightarrow R_{eq} = \frac{2R_1 R_2}{n(R_1 + R_2)}$$

$$\Rightarrow \frac{d}{K_{eq} nA} = \frac{2}{n} \cdot \left(\frac{\frac{d}{K_1 A} \cdot \frac{d}{K_2 A}}{\frac{d}{K_1 A} + \frac{d}{K_2 A}}\right)$$

$$\frac{1}{K_{eq} A} = \frac{2\left(\frac{1}{K_1 A} \cdot \frac{1}{K_2 A}\right)}{\frac{K_2 A + K_1 A}{K_1 K_2 A^2}}$$

$$\frac{1}{K_{eq} A} = \frac{2}{K_1 A + K_2 A}$$

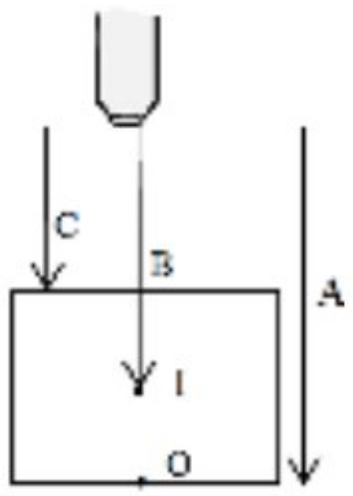
$$\Rightarrow \boxed{K_{eq} = \frac{K_1 + K_2}{2}}$$

43) $\delta = i + e - A$

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$\Rightarrow A = i + e - \delta = (32^\circ + 58^\circ) - 30^\circ = 60^\circ$

44)



$$\text{RI of slab} = \frac{\text{True thickness}}{\text{Apparent thickness}} = \frac{A-C}{B-C}$$

$$\Rightarrow RI = \frac{(8.140-4.900)}{(6.925-4.900)} = \frac{3.24}{2.025} = 1.6$$

45) Let, t = actual thickness of slab; shift = x = 6 cm

$$\text{shift} = x = t\left(1-\frac{1}{\mu}\right)$$

$$\Rightarrow 6 = t\left(1-\frac{1}{1.5}\right)$$

$\Rightarrow t = 6 \times 3 = 18$ cm

CHEMISTRY

46)

Fact Based

47) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

V_1 = Volume of N_2 gas

P_1 = pressure of N_2- Aqueous tension

T_1 = Room temperature

$$V_2 = \frac{P_1V_1 \times 273}{T_1 \times 760}$$

$$V_2 = \frac{700}{300} \times \frac{273}{760}$$

= 33.53 ml

$$\%N = \frac{28}{22400} \times \frac{33.53}{0.25} \times 100$$

= 16.76

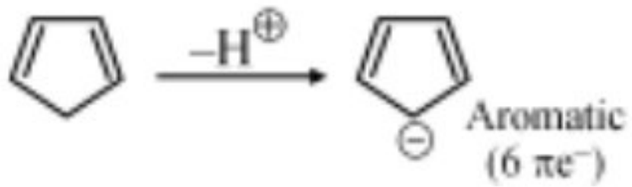
48) While testing for halogens, lassaigne's extract is boiled with conc. HNO_3 to decompose Na_2S and NaCN if formed.

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49) Thin layer chromatography – Example of adsorption chromatography

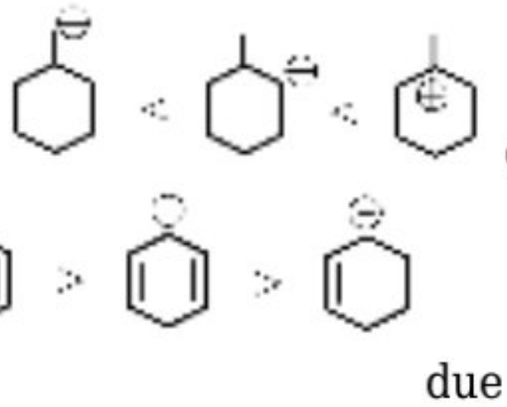
50) Both statement I & statement II are correct.

51) Kjeldahl method can not be used to estimate nitro compound azo compound N in ring because these compounds do not form ammonium sulphate with conc H_2SO_4.

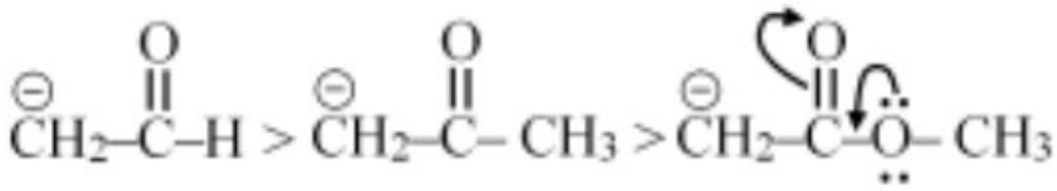
52) 

53) $HC \equiv CH + NaOH \rightleftharpoons HC \equiv CNa + H_2O$
(WA) (SA)

54) $-\ddot{\underset{\cdot\cdot}{O}}-CH=CH_2$
(□p)
effect are + M >> -I.

55) 
due to +H effect

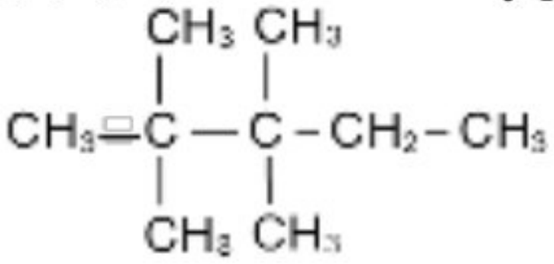
due to resonance



56)

For anti-aromaticity $4n\pi e^-$ (n = 1,2,3....) should be present.
I & IV are aromatic ($6\pi e^-s$)
II & III are non aromatic due to nonplanarity.

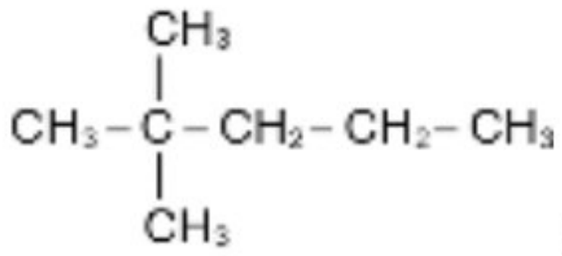
57) **Solution :**
An isopropyl group is a group with the structure $-CH(CH_3)_2$
To determine which compound has one isopropyl group. We will analyze each option.
(1) 2, 2, 3, 3-Tetramethylpentane.



(No isopropyl group)

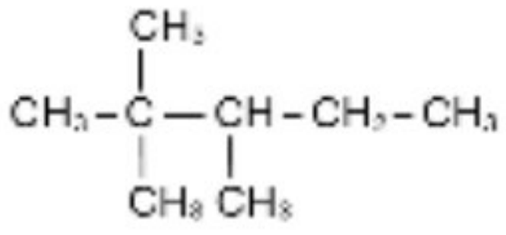
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(2) 2, 2-Dimethylpentane



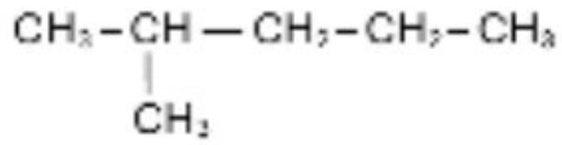
(No isopropyl group)

(3) 2,2,3-Trimethylpentane



(No isopropyl group)

(4) 2-Methylpentane

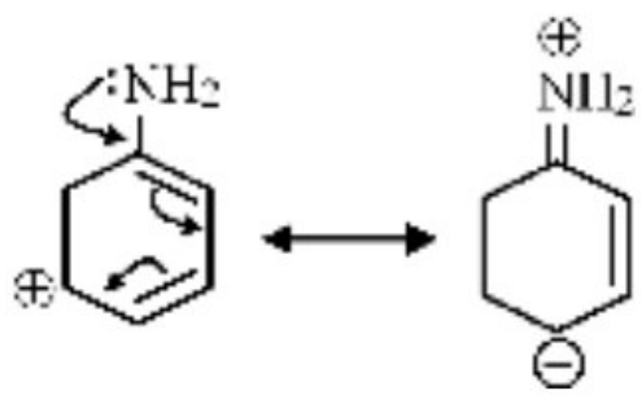


This compound has (Isopropyl group)

Hence,

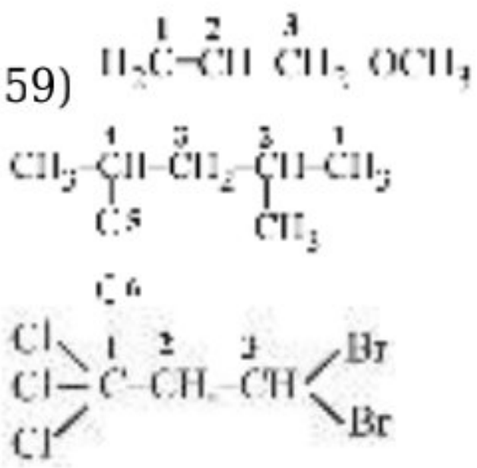
Option (4) is correct : 2-methyl pentane

58)

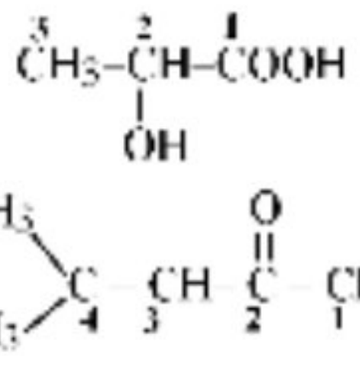


The + M effect of NH_2 is stabilizing the carbocation.

59)

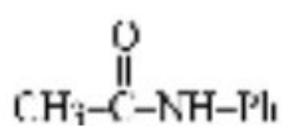


60)



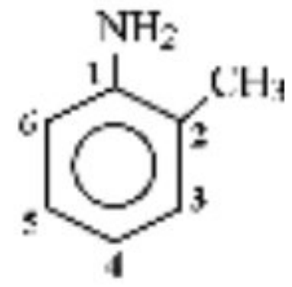
(Lactic acid)

(Mesityl oxide)

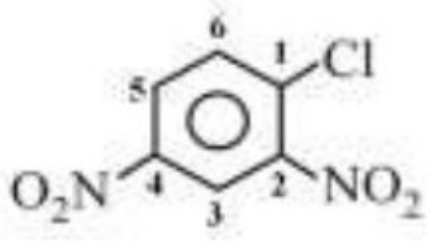


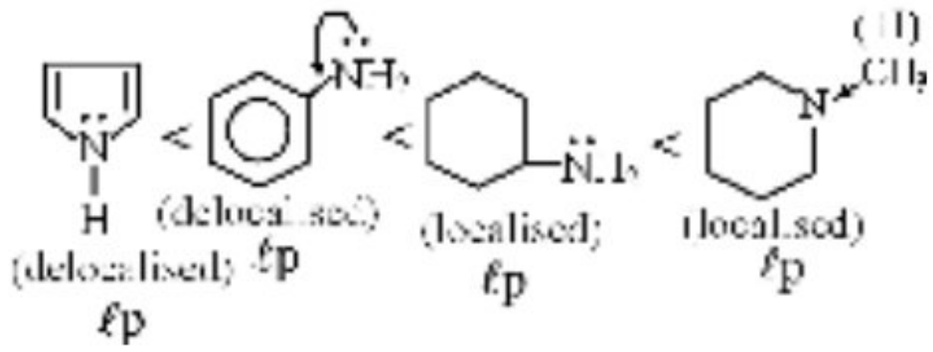
(Acetanilide)

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61) 

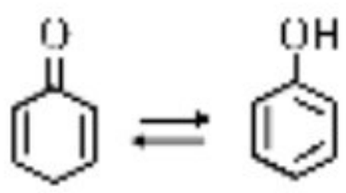
(o-Toluidine)



62) 

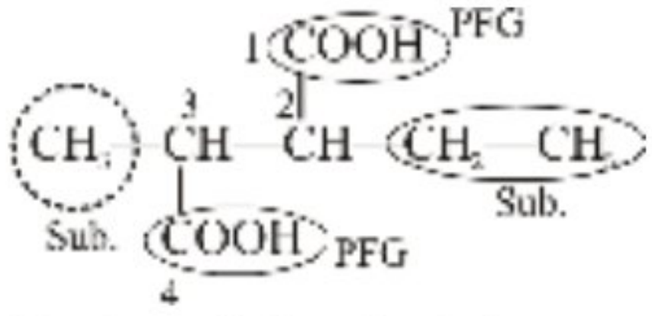
63)

All the members of Homologous series have same chemical properties as they have same functional group.

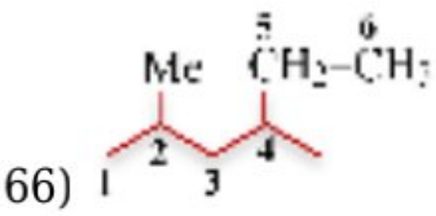
64) 

Phenol
(Aromatic)

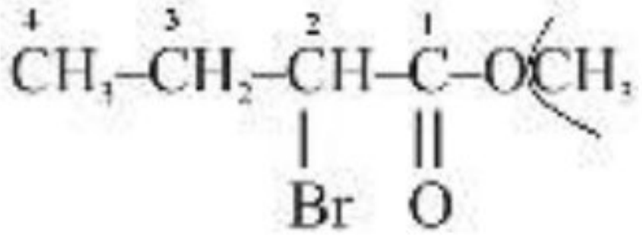
65)

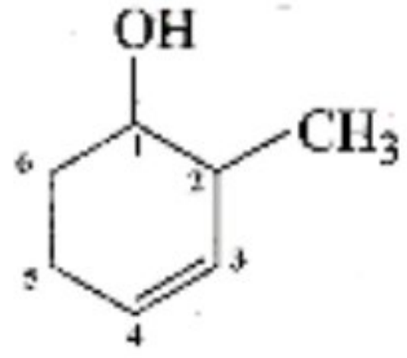


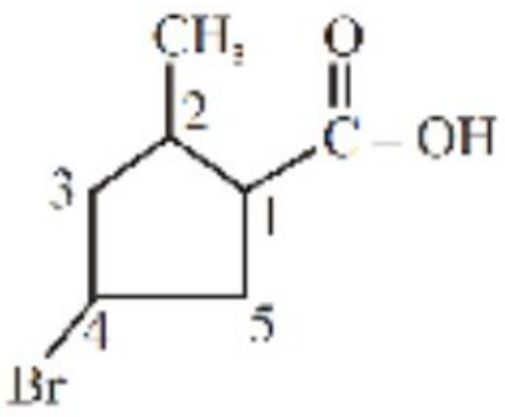
* Both methyl and ethyl are at equivalent position so ethyl gets less locant.

66) 

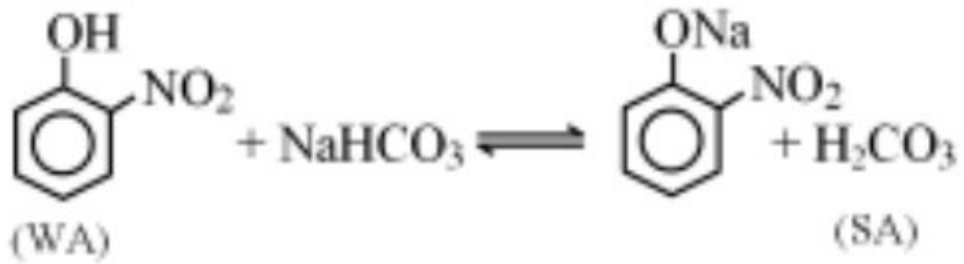
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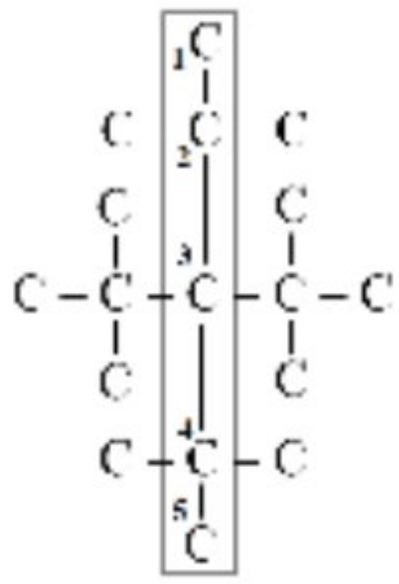
67) 

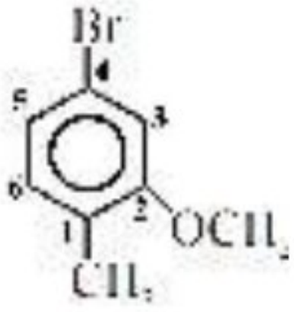
68) 

69) 

4-bromo-2-methyl cyclopentane carboxylic Acid

70) 

71) 

72) 

(Minimum number rule)

73)

• The assertion states that D-configuration implies dextrorotatory. This is false. D and L

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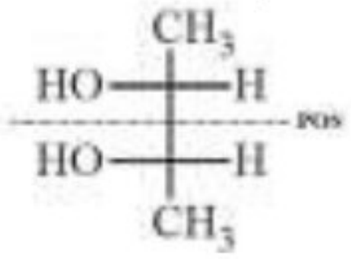
configurations are related to the arrangement of atoms in space, while dextrorotatory and levorotatory are related to how a compound interacts with plane-polarized light.
• The reason states that D-configuration implies clockwise rotation of plane-polarized light. This is also false. The D/L system does not predict the direction of rotation.

74) If POS & COS are absent then molecule, optically active.

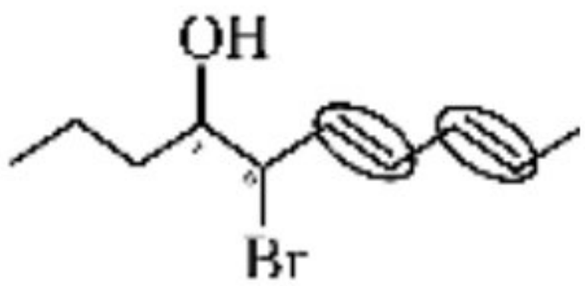
75)

Structure (II) is optically inactive due to presence of POS.

76) ⇒ If molecule has symmetry element POS/COS then molecule so said to be Optically



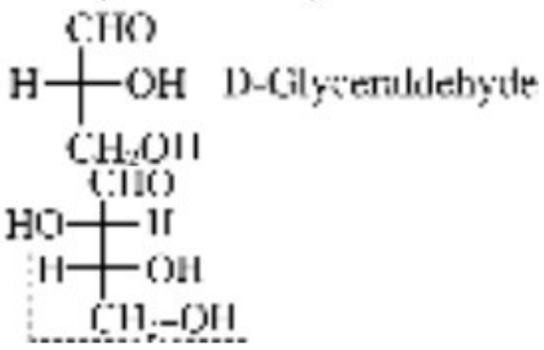
inactive. □
to optically inactive due to presence of POS.



77) = unsymmetrical compound
No of system across which stereoisomers are possible n = 4
No of stereoisomers $2^4 = 16$.

78)

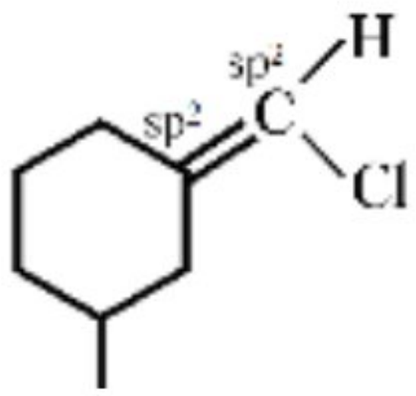
For D-form of compound, it is derivative of
D-Glyceraldehyde.



79)

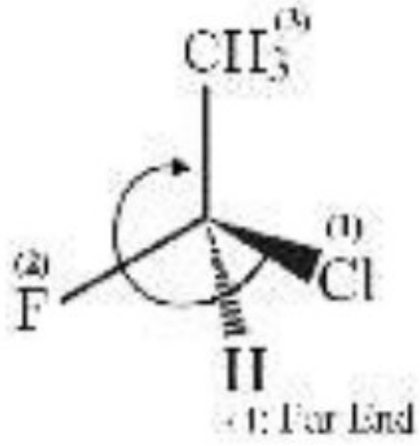
I > I' (due to complete octet)
II' > II (due to complete octet)

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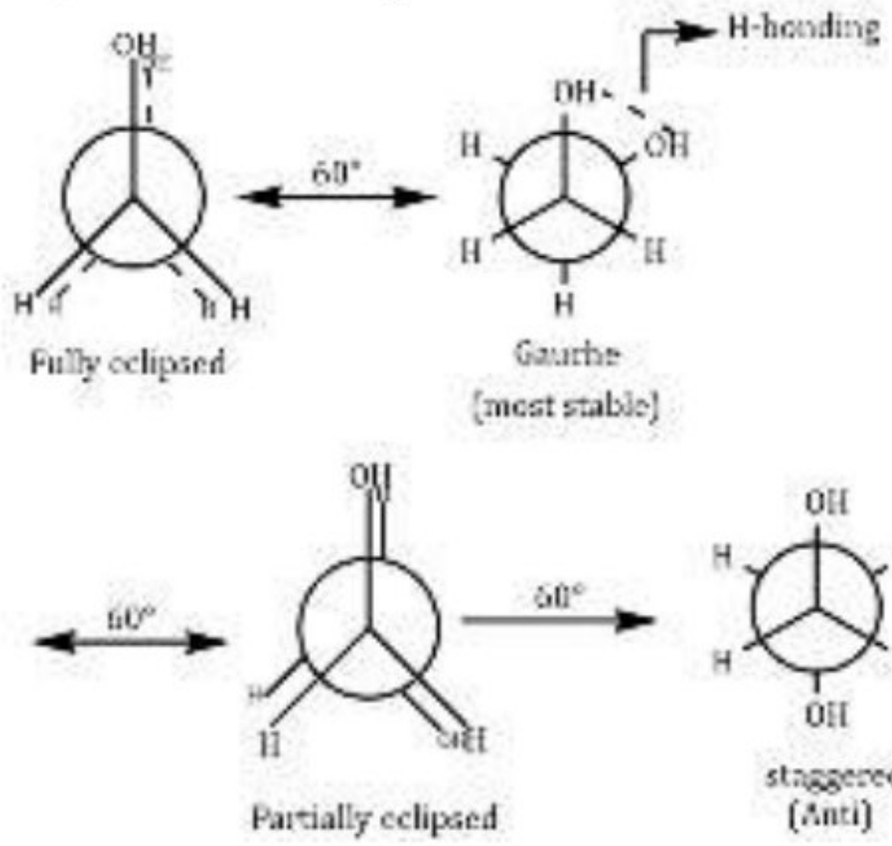
80)

Both sp^2 carbon have different valencies.



81)

82) **D Answer Explanation:**

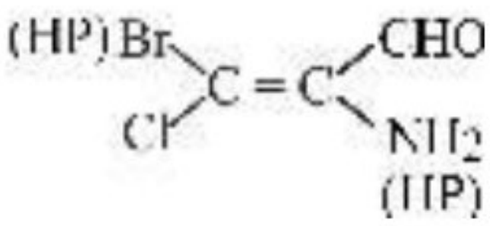


83)

For GI restricted rotation along C = C | C = N | N = N cyclo alkane | Cyclo alkene.

84)

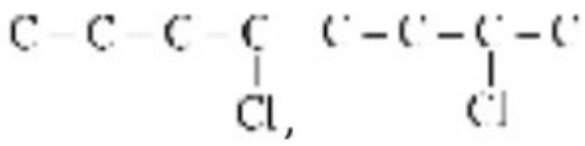
NCERT 11th Page No. 305



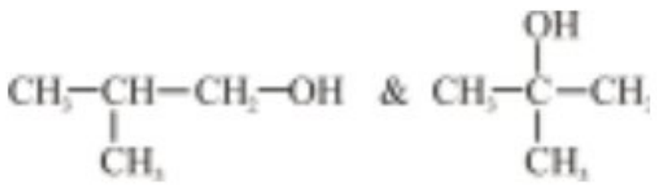
85)

E-Configuration

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86) 

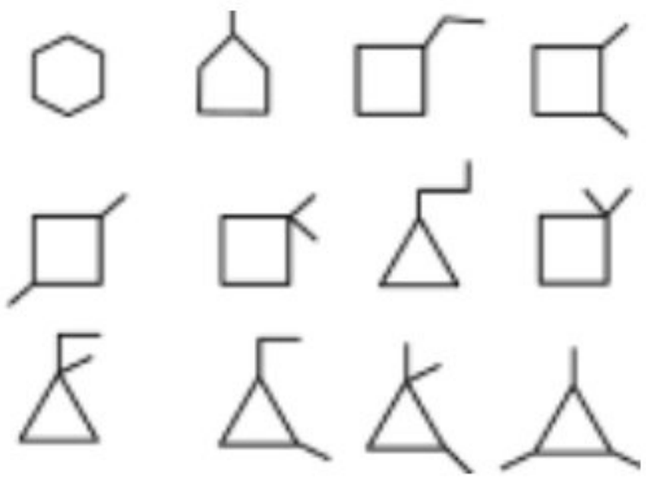
87)



are position isomers.

88)

For metamers polyvalent FG should be present.

89) 

90) 

Does not have any α – H.

BIOLOGY

91) NCERT XII, Page # 205

92) NCERT XII, Page # 205

93) NCERT XII, Page # 205

94) NCERT XII, Page # 206

95) NCERT XII, Page # 207

96) NCERT XII, Page # 207

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97) NCERT XII, Page # 207

98) NCERT XII, Page # 207

99) NCERT XII, Page # 208

100) NCERT XII, Page # 209

101) NCERT XII, Page # 207
10 percent of GPP is stored as NPP.

102) NCERT XII, Page # 212
Hint : Sparrow, Cockroach.

103) NCERT XII, Page # 212

104) NCERT XII, Page # 213

105) NCERT XlI, Page # 213
Omnivores – Human sparrow

106) NCERT XII, Page # 210

107) NCERT XII, Page # 210

108) NCERT XII, Page # 212, 213

109) NCERT XII, Page # 213

110) NCERT XII, Page # 212-213

111) Only (i) is correct
NCERT XII, Page # 211

112) NCERT-XII, Page # 213

113) NCERT-XII, Pg. No. # 222

114) NCERT-XII, Pg. No. # 220

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115) NCERT-XII, Pg. No. # 220

116) NCERT-XII, Pg. No. # 224

117) NCERT-XII, Pg. No. # 220

118) NCERT-XII, Pg. No. # 224

119) NCERT-XII, Pg. No. # 221

120) NCERT-XII, Pg. No. # 224

121) NCERT-XII, Pg. No. # 218

122) NCERT-XII, Pg. No. # 225

123) NCERT-XII, Pg. No. # 218

124) NCERT-XII, Pg. No. # 216, 223

125) NCERT-XII, Pg. No. # 218

126) NCERT-XII, Pg. No. # 218

127) NCERT-XII, Pg. No. # 221

128) NCERT-XII Pg. No. # 222

129) NCERT-XII Pg. No. # 218, 219

130) NCERT-XII Pg. No. # 221, 222, 223, 225

131) NCERT-XII Pg. No. # 217, 220-221

132) NCERT-XII, Pg. No. # 223

133) NCERT-XII, Pg. No. # 220

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138) NCERTPage No. # 242-243
The exopthalmic goitre is a form of hyperthyroidism, characterised by enlargement of the thyroid gland, protrusion of eye balls, increased BMR and weight loss, also called grave's disease.
एक्सोप्थोल्मिक गोइटर हाइपर थायडिज्म का एक रूप हैं, जो थाइरॉइड ग्रन्थी के बढने, नेत्र गोलक के बाहर निकलने, BMR में वृद्धि और वजन घटाने की विशेषता है जिसे ग्रेब्स रोग भी कहा जाता है।

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